

Phase 2: Fields, Galois Theory, and Frobenius

A detailed self-study workbook for the Chebotarev Density Theorem course

Field extensions, separability, normality, splitting fields, Galois correspondence, cyclotomic fields, finite fields, and the finite-field Frobenius

Design principle. Every central definition is paired with (i) the problem it solves, (ii) a mental model, (iii) examples and near-misses, and (iv) a derivation showing how it participates in later mathematics. The goal is not to recognize Galois-theory vocabulary. It is to be able to reconstruct the vocabulary and the main arguments when the book is closed.

How to use this workbook

Phase 2 has a very specific job in the Chebotarev project: it must make the phrase

“the Frobenius conjugacy class in $\text{Gal}(L/K)$ ”

feel inevitable rather than mysterious. To reach that point you need to own the language of field extensions and understand why automorphisms, conjugates, normality, separability, permutation actions, and finite-field Frobenius fit together.

This workbook therefore uses a four-pass method.

1. **Recall before recognition.** Cover a definition and try to reconstruct it before reading it.
2. **Derive the definition’s shape.** Ask what would fail if one clause were removed.
3. **Move between three representations.** For every important example, move between a field description, a polynomial description, and a group/permutation description.
4. **Reconstruct proofs.** After reading a proof, close the book and write the dependency chain before reproducing details.

Memory hook

A reliable test of Galois-theory understanding is whether you can move in both directions:

polynomial roots $\longleftrightarrow$ field automorphisms $\longleftrightarrow$ permutations/group structure.

If one arrow feels one-way only, that is the place to repair.

Recommended pacing

At 8–10 focused hours per week, budget about four weeks:

Week 1	Chapters 1–3: extensions, algebraic elements, minimal polynomials, embeddings, separability.
Week 2	Chapters 4–5: normality, splitting fields, automorphism groups, the fundamental theorem of Galois theory.
Week 3	Chapters 6–7: explicit Galois groups and cyclotomic fields.
Week 4	Chapters 8–10: finite fields, Frobenius, factorization/cycle type, synthesis and final exam.

The calendar is not the progression rule. The mastery gates are.

What “ready for Phase 3” means

You should be able to do four things without constant reference to notes:

1. state and motivate the central definitions;
2. prove the main structural results at the level indicated in this workbook;
3. compute several representative examples, including a quadratic, a biquadratic, a cubic splitting field, a cyclotomic field, and a finite field extension;
4. explain why Frobenius is canonical in a finite field and why conjugacy appears when a group acts on roots without a preferred labeling.

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Chapter 1

Bridge diagnostic: what Phase 1 should have left you with

The purpose of this chapter is not to re-examine all of commutative algebra. It checks only the algebraic reflexes Phase 2 will use repeatedly.

1.1 Closed-book diagnostic

For each problem mark **C** (confident), **U** (uncertain), or **N** (no workable approach). Do not read the solutions until you have attempted all of them.

Exercise 1.1. Let F be a field and $f \in F[x]$. Prove that $F[x]/(f)$ is a field if and only if f is irreducible.

Exercise 1.2. Explain, both formally and intuitively, why maximal ideals are exactly those ideals $\mathfrak{m}$ for which $R/\mathfrak{m}$ is a field.

Exercise 1.3. Solve the simultaneous congruences

$$x \equiv 2 \pmod{5}, \quad x \equiv 3 \pmod{7},$$

and state the Chinese remainder theorem in the form that explains why a unique solution class modulo 35 exists.

Exercise 1.4. Let R be an integral domain and $\mathfrak{p}$ a prime ideal. State the definition of $R_{\mathfrak{p}}$ and explain which elements become units.

Exercise 1.5. State what it means for α to be integral over a ring A . Then explain why every element algebraic over a field K is automatically integral over K .

Exercise 1.6. Let V be a vector space over K with basis $v_1, \dots, v_n$. Explain why a K -linear map $T : V \rightarrow V$ is completely determined by $T(v_1), \dots, T(v_n)$, and why this simple fact will later bound the number of field automorphisms.

1.2 Interpretation

If Exercises 1.1–1.5 are largely inaccessible, repair the relevant Phase 1 sections before continuing. Exercise 1.6 is different: if it feels rusty, simply review finite-dimensional linear algebra alongside Chapter 2.

Mastery gate

Proceed if you can use quotient rings, maximal ideals, and polynomial quotients comfortably. You do *not* need the full theory of Dedekind domains in active memory yet; Phase 3 will reactivate it in number-field form.

Chapter 2

Fields and field extensions: enlarging the universe of scalars

2.1 What a field extension really is

Definition 2.1 (Field extension). A *field extension* L/K means that K is a subfield of L . We also write $L \supseteq K$ or L/K .

The notation L/K does not denote a quotient. It denotes a relationship: L is a larger field in which the elements of K already live.

Why this definition?

Why insist that K be a *subfield*, not merely a subset? Because we want every arithmetic operation and every scalar relation already valid in K to remain valid in L . Once $K \subseteq L$ is a subfield, L is automatically a vector space over K : scalar multiplication is just multiplication in L .

Example 2.2. $\mathbb{C}/\mathbb{R}$, $\mathbb{R}/\mathbb{Q}$, $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$, and $\mathbb{F}_{p^n}/\mathbb{F}_p$ are field extensions. By contrast, $\mathbb{Z} \subseteq \mathbb{Q}$ is not a field extension because $\mathbb{Z}$ is not a field.

2.2 Degree: measuring how much larger the field is

Definition 2.3 (Degree). The *degree* of L/K is the dimension of L as a vector space over K :

$$[L : K] := \dim_K L.$$

If this dimension is finite, L/K is a *finite extension*.

Intuition

A field extension introduces new arithmetic elements, but vector-space dimension measures how many independent “directions” of new arithmetic are needed. In $\mathbb{Q}(\sqrt{2})$ every element is $a + b\sqrt{2}$, so only the directions 1 and $\sqrt{2}$ are needed; hence the degree is 2.

Worked example

Show that $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$.

Every rational expression in $\sqrt{2}$ reduces to $a + b\sqrt{2}$. Indeed,

$$\frac{a + b\sqrt{2}}{c + d\sqrt{2}} = \frac{(a + b\sqrt{2})(c - d\sqrt{2})}{c^2 - 2d^2},$$

provided $c + d\sqrt{2} \neq 0$. Thus $\mathbb{Q}(\sqrt{2})$ is spanned by $1, \sqrt{2}$ over $\mathbb{Q}$. These vectors are linearly independent: if $a + b\sqrt{2} = 0$ with $a, b \in \mathbb{Q}$ and $b \neq 0$, then $\sqrt{2} = -a/b \in \mathbb{Q}$, contradiction. Therefore $\{1, \sqrt{2}\}$ is a basis.

2.3 The tower law

Suppose $K \subseteq E \subseteq L$. Degree should multiply, just as dimensions multiply when one changes scalar fields.

Theorem 2.4 (Tower law). *If $[L : E]$ and $[E : K]$ are finite, then*

$$[L : K] = [L : E][E : K].$$

Proof. Choose an E -basis $u_1, \dots, u_m$ of L and a K -basis $v_1, \dots, v_n$ of E . We claim that the mn products $u_i v_j$ form a K -basis of L .

First, they span. Given $x \in L$, write $x = \sum_i e_i u_i$ with $e_i \in E$. Then each $e_i = \sum_j a_{ij} v_j$ with $a_{ij} \in K$. Hence

$$x = \sum_{i,j} a_{ij} u_i v_j.$$

Second, they are independent. Suppose

$$\sum_{i,j} a_{ij} u_i v_j = 0.$$

Regroup by i :

$$\sum_i \left(\sum_j a_{ij} v_j \right) u_i = 0.$$

The u_i are E -linearly independent, so for every i ,

$$\sum_j a_{ij} v_j = 0.$$

The v_j are K -linearly independent, so all $a_{ij} = 0$. Thus the products form a basis and the degree is mn . $\square$

Memory hook

The tower law is a divisibility machine. If $K \subseteq E \subseteq L$ and $[L : K]$ is prime, then there is no proper intermediate field: both factors in $[L : K] = [L : E][E : K]$ are positive integers.

2.4 Adjoining elements

Definition 2.5. For L/K and $\alpha \in L$:

- $K[\alpha]$ is the smallest subring of L containing K and α ;
- $K(\alpha)$ is the smallest subfield of L containing K and α .

Every element of $K[\alpha]$ is a polynomial in α :

$$K[\alpha] = \{f(\alpha) : f \in K[x]\}.$$

But $K(\alpha)$ allows quotients of such expressions. These are equal precisely in the important algebraic case, as we now derive.

2.5 Algebraic elements and minimal polynomials

Definition 2.6 (Algebraic and transcendental). An element $\alpha \in L$ is *algebraic over K* if some nonzero polynomial $f \in K[x]$ satisfies $f(\alpha) = 0$. Otherwise α is *transcendental over K* .

Why this definition?

The distinction asks whether the new element obeys a finite algebraic relation over the old field. An algebraic element can therefore be compressed into finitely many powers; a transcendental element behaves like a genuinely free indeterminate.

Consider the evaluation homomorphism

$$\text{ev}_\alpha : K[x] \longrightarrow L, \quad f \longmapsto f(\alpha).$$

Its image is $K[\alpha]$. If α is algebraic, the kernel is a nonzero ideal. Since $K[x]$ is a PID, the kernel is generated by a single monic polynomial.

Definition 2.7 (Minimal polynomial). If α is algebraic over K , its *minimal polynomial* $m_{\alpha,K}(x)$ is the unique monic generator of

$$\ker(\text{ev}_\alpha).$$

Equivalently, it is the unique monic polynomial of least positive degree in $K[x]$ having α as a root.

Proposition 2.8. *The minimal polynomial $m_{\alpha,K}$ is irreducible in $K[x]$.*

Proof. Suppose $m = fg$ with $f, g \in K[x]$ both of positive degree. Evaluating at α gives

$$0 = m(\alpha) = f(\alpha)g(\alpha).$$

Because L is a field, it has no zero divisors, so $f(\alpha) = 0$ or $g(\alpha) = 0$. But then a polynomial of smaller positive degree than m lies in the kernel, contradicting minimality. $\square$

Theorem 2.9 (Simple algebraic extensions). *If α is algebraic over K with minimal polynomial m , then*

$$K[\alpha] \cong K[x]/(m).$$

Moreover $K[\alpha]$ is a field, hence

$$K[\alpha] = K(\alpha),$$

and

$$[K(\alpha) : K] = \deg m.$$

Proof. The first isomorphism theorem applied to ev_α gives

$$K[x]/\ker(\text{ev}_\alpha) \cong \text{im}(\text{ev}_\alpha) = K[\alpha].$$

Since the kernel is (m) , this gives the first claim. By Proposition 2.8, m is irreducible, so (m) is maximal in $K[x]$; therefore $K[x]/(m)$ is a field. Thus $K[\alpha]$ itself is a field containing K and α , so by minimality of $K(\alpha)$ we have $K(\alpha) \subseteq K[\alpha]$. The reverse inclusion is automatic, hence equality.

Finally, Euclidean division gives every residue class modulo m a unique representative of degree $< n := \deg m$. Thus

$$1, \alpha, \dots, \alpha^{n-1}$$

span $K(\alpha)$ over K . They are independent, because a relation of degree $< n$ would produce a nonzero polynomial of degree smaller than m annihilating α . Therefore they form a basis. $\square$

Intuition

The relation $m(\alpha) = 0$ is a reduction rule. If m has degree n , every power α^r with $r \geq n$ can be rewritten using lower powers. This is why algebraicity produces finite-dimensional extensions.

2.6 Irreducibility tools used later

The minimal polynomial is defined by irreducibility, so we need a few reliable ways to prove a polynomial irreducible. The following tools will also be used in the cyclotomic chapter.

Definition 2.10 (Primitive integer polynomial). A nonzero polynomial in $\mathbb{Z}[x]$ is *primitive* if the greatest common divisor of its coefficients is 1.

Lemma 2.11 (Gauss's lemma). *The product of primitive polynomials in $\mathbb{Z}[x]$ is primitive. Consequently, a primitive polynomial $f \in \mathbb{Z}[x]$ is irreducible in $\mathbb{Z}[x]$ if and only if it is irreducible in $\mathbb{Q}[x]$.*

Proof. Suppose primitive $f, g \in \mathbb{Z}[x]$ had a product all of whose coefficients were divisible by a prime p . Reducing coefficients modulo p would give

$$\bar{f} \bar{g} = 0 \quad \text{in } \mathbb{F}_p[x].$$

But $\mathbb{F}_p[x]$ is an integral domain, so $\bar{f} = 0$ or $\bar{g} = 0$, meaning p divides every coefficient of one factor, contradicting primitivity. Thus fg is primitive.

Now suppose a primitive $f \in \mathbb{Z}[x]$ factors nontrivially in $\mathbb{Q}[x]$. Clear denominators in the two rational factors and divide each resulting integer polynomial by the gcd of its coefficients. We obtain primitive $g_0, h_0 \in \mathbb{Z}[x]$ and a rational number $c \neq 0$ such that

$$f = cg_0h_0.$$

The product g_0h_0 is primitive by the first part. Comparing contents (the gcd of coefficients, normalized positive) forces $|c| = 1$ because f is also primitive. Hence, up to a sign, f factors in $\mathbb{Z}[x]$. The reverse implication is immediate. $\square$

Proposition 2.12 (Rational-root test). *If*

$$f(x) = a_n x^n + \dots + a_0 \in \mathbb{Z}[x]$$

and a reduced fraction r/s is a rational root, then $r \mid a_0$ and $s \mid a_n$.

Proof. Substitute $x = r/s$ and multiply by s^n :

$$a_n r^n + a_{n-1} r^{n-1} s + \dots + a_0 s^n = 0.$$

All terms except the first are divisible by s , so $s \mid a_n r^n$. Since $\gcd(r, s) = 1$, $s \mid a_n$. Similarly, isolating the last term shows $r \mid a_0 s^n$, and coprimality gives $r \mid a_0$. $\square$

Theorem 2.13 (Eisenstein criterion). *Let*

$$f(x) = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$$

be primitive. If a prime p satisfies

$$p \nmid a_n, \quad p \mid a_i \ (0 \leq i < n), \quad p^2 \nmid a_0,$$

then f is irreducible over $\mathbb{Q}$.

Proof. By Gauss's lemma it suffices to rule out a factorization $f = gh$ with nonconstant $g, h \in \mathbb{Z}[x]$. Reduce modulo p . All nonleading coefficients of f vanish, so

$$\bar{f} = \bar{a}_n x^n.$$

Since $\mathbb{F}_p[x]$ is a UFD, $\bar{g}$ and $\bar{h}$ must each be a nonzero scalar times a power of x . Because both factors have positive degree, their constant terms are divisible by p . Therefore the constant term

$$a_0 = g(0)h(0)$$

is divisible by p^2 , contradicting the hypothesis. $\square$

Memory hook

For a cubic, the rational-root test is often enough: a reducible cubic over a field has a linear factor. Eisenstein is especially useful for $x^n - p$ -type polynomials. Gauss's lemma is the bridge that lets integer divisibility arguments prove irreducibility over $\mathbb{Q}$.

2.7 Finite extensions are algebraic

Proposition 2.14. *If $[L : K] = n < \infty$, then every $\alpha \in L$ is algebraic over K , with minimal polynomial degree at most n .*

Proof. The $n + 1$ vectors

$$1, \alpha, \alpha^2, \dots, \alpha^n$$

lie in the n -dimensional K -vector space L , so they are linearly dependent. Thus there exist $a_0, \dots, a_n \in K$, not all zero, such that

$$a_0 + a_1 \alpha + \cdots + a_n \alpha^n = 0.$$

Hence α is algebraic. Its minimal polynomial divides any annihilating polynomial, so its degree is at most n . $\square$

The converse is false if infinitely many algebraic elements are adjoined. For example $\overline{\mathbb{Q}}/\mathbb{Q}$ is algebraic but infinite-degree.

2.8 Several generators

Write

$$K(\alpha_1, \dots, \alpha_r)$$

for the smallest subfield containing all generators. Build it as a tower:

$$K \subseteq K(\alpha_1) \subseteq K(\alpha_1, \alpha_2) \subseteq \cdots.$$

If each α_i is algebraic over the field generated before it, the whole extension is finite by the tower law.

Worked example

Compute $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}]$.

Certainly $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$. Over $\mathbb{Q}(\sqrt{2})$, the element $\sqrt{3}$ has degree at most 2. It cannot have degree 1, because that would mean $\sqrt{3} = a + b\sqrt{2}$ with $a, b \in \mathbb{Q}$. Squaring gives

$$3 = a^2 + 2b^2 + 2ab\sqrt{2}.$$

Thus $2ab = 0$. If $a = 0$, then $3 = 2b^2$, impossible for rational b ; if $b = 0$, then $3 = a^2$, also impossible. Hence $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$, so the second degree is 2. Therefore

$$[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = 4.$$

A basis is $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$.

2.9 Definition recall drill

Before looking back, state and motivate:

1. field extension and extension degree;
2. algebraic versus transcendental;
3. $K[\alpha]$ versus $K(\alpha)$;
4. minimal polynomial;
5. why $K[\alpha] = K(\alpha)$ for algebraic α ;
6. the tower law and one consequence.

Then give one example and one nonexample for each distinction that admits one.

2.10 Exercises

Exercise 2.1. Suppose $[L : K] = p$ is prime. Prove that there is no intermediate field E with $K \subsetneq E \subsetneq L$.

Exercise 2.2. Find the minimal polynomial over $\mathbb{Q}$ of $\sqrt{2} + \sqrt{3}$, and compute $[\mathbb{Q}(\sqrt{2} + \sqrt{3}) : \mathbb{Q}]$.

Exercise 2.3. Let α be algebraic over K and nonzero. Starting from an equation satisfied by α , explicitly show that $\alpha^{-1} \in K[\alpha]$.

Exercise 2.4. Suppose α and β are algebraic over K . Prove that $K(\alpha, \beta)/K$ is finite and deduce that $\alpha + \beta$, $\alpha\beta$, and (if $\beta \neq 0$) α/β are algebraic over K .

Exercise 2.5. Let $\alpha \in L$ be algebraic over K , where $[L : K] < \infty$. Prove that $[K(\alpha) : K]$ divides $[L : K]$.

Exercise 2.6. Prove that $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$. Describe every element uniquely in the form $a + b\sqrt[3]{2} + c\sqrt[3]{4}$.

Exercise 2.7. Let t be transcendental over K . Prove that $K[t] \cong K[x]$ but $K[t] \neq K(t)$. Give an explicit element of $K(t)$ not lying in $K[t]$.

Exercise 2.8. Suppose $[L : K] = 4$. What degrees can an intermediate field E have over K ? Does the tower law guarantee that an intermediate field of degree 2 exists? Explain.

Exercise 2.9. Prove directly that $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$ are linearly independent over $\mathbb{Q}$.

Exercise 2.10. Let α be algebraic over K . Prove that every $f \in K[x]$ satisfying $f(\alpha) = 0$ is divisible by $m_{\alpha, K}$.

Chapter 3

Embeddings, conjugates, and separability

A field extension becomes Galois-theoretic when we stop asking only which elements it contains and begin asking how many structure-preserving ways it can sit inside a larger field.

3.1 Field embeddings

Definition 3.1 (Field embedding). A K -embedding $\sigma : L \hookrightarrow M$ is an injective field homomorphism satisfying $\sigma(a) = a$ for all $a \in K$.

Any field homomorphism is automatically injective unless it is the zero map, because its kernel is an ideal of a field. Since a unital field homomorphism sends 1 to 1, it is not zero.

Why this definition?

A K -embedding is allowed to move the newly adjoined elements but must keep the base field fixed. That is exactly what it means to regard two algebraic realizations as the same “over K .”

3.2 An embedding is determined by the image of a generator

Let $L = K(\alpha)$ with α algebraic. If $\sigma : L \hookrightarrow M$ is a K -embedding, then for any polynomial $f \in K[x]$,

$$\sigma(f(\alpha)) = f(\sigma(\alpha)).$$

Thus σ is determined by $\sigma(\alpha)$. Moreover, applying σ to

$$m_{\alpha,K}(\alpha) = 0$$

gives

$$m_{\alpha,K}(\sigma(\alpha)) = 0.$$

So the image of α must be another root of its minimal polynomial.

Proposition 3.2 (Embeddings of a simple extension). Let $L = K(\alpha)$ and let M be a field containing K . For each root $\beta \in M$ of $m_{\alpha,K}$ there is a unique K -embedding

$$\sigma_\beta : K(\alpha) \hookrightarrow M$$

with $\sigma_\beta(\alpha) = \beta$.

Proof. Use the quotient presentation $K(\alpha) \cong K[x]/(m)$. Evaluation at β defines a homomorphism $K[x] \rightarrow M$ whose kernel contains (m) because $m(\beta) = 0$. Hence it descends to

$$K[x]/(m) \longrightarrow M,$$

sending the class of x to β . Because the domain is a field, this nonzero homomorphism is injective. Uniqueness follows because every element is a polynomial in α of degree $< \deg m$. $\square$

Intuition

Conjugate roots are not merely visually similar roots of the same equation. They are precisely the possible destinations of the generator under embeddings that refuse to move the base field.

Theorem 3.3 (Extension of embeddings for finite algebraic extensions). *Let E/K be a finite algebraic extension, let Ω be algebraically closed, and let $\sigma : K \hookrightarrow \Omega$ be an embedding. Then σ extends to an embedding*

$$\tilde{\sigma} : E \hookrightarrow \Omega.$$

More generally, if $K \subseteq E_0 \subseteq E$ and an embedding $E_0 \hookrightarrow \Omega$ is already given, it extends to E .

Proof. Because E/E_0 is finite, choose generators

$$E = E_0(\alpha_1, \dots, \alpha_r).$$

We extend one generator at a time. Suppose an embedding

$$\sigma_i : E_0(\alpha_1, \dots, \alpha_i) \hookrightarrow \Omega$$

has been constructed. Let $m_i(x)$ be the minimal polynomial of α_{i+1} over the current source field. Apply σ_i to its coefficients to obtain a polynomial $m_i^{\sigma_i}(x)$ over the image field. Since Ω is algebraically closed, choose a root β of $m_i^{\sigma_i}$. The quotient argument used in Proposition 3.2 then extends σ_i uniquely after the choice of β by sending α_{i+1} to β . Repeating finitely many times produces the desired embedding of E . $\square$

Memory hook

To extend an embedding across an algebraic generator, transport its minimal polynomial and choose one of its roots. Galois theory repeatedly uses this single move.

3.3 Repeated roots and the derivative criterion

Lemma 3.4. *Let $f \in K[x]$ and let α lie in an extension field. Then α is a repeated root of f if and only if*

$$f(\alpha) = f'(\alpha) = 0.$$

Proof. If $f = (x - \alpha)^2 g$, differentiation gives

$$f' = 2(x - \alpha)g + (x - \alpha)^2 g',$$

so both vanish at α .

Conversely, write $f = (x - \alpha)g$. Differentiating gives

$$f' = g + (x - \alpha)g'.$$

Thus $f'(\alpha) = g(\alpha)$. If this is zero, then $x - \alpha$ divides g , so $(x - \alpha)^2$ divides f . $\square$

Definition 3.5 (Separable polynomial and element). A polynomial is *separable* if it has no repeated roots in a splitting field. An algebraic element α is *separable over K* if its minimal polynomial over K is separable. An algebraic extension L/K is separable if every element of L is separable over K .

Why this definition?

Why care about repeated roots? Because distinct roots correspond to distinct embeddings. Repeated roots reduce the number of genuinely different images available to an algebraic element. Separability is exactly the condition that the polynomial relation has as many distinct geometric realizations as its degree permits.

Proposition 3.6 (Derivative criterion). A nonconstant polynomial $f \in K[x]$ is separable if and only if

$$\gcd(f, f') = 1.$$

If f is irreducible, then it is inseparable if and only if $f' = 0$.

Proof. By Lemma 3.4, f has a repeated root exactly when f and f' have a common root in a splitting field. This occurs exactly when they have a nonconstant common factor, i.e. $\gcd(f, f') \neq 1$.

If f is irreducible, then $\gcd(f, f')$ is either 1 or an associate of f . But $\deg f' < \deg f$ unless $f' = 0$. Therefore the latter case is possible only if $f' = 0$. $\square$

3.4 Characteristic zero and positive characteristic

If $\text{char } K = 0$ and f is nonconstant, f' can vanish only if f is constant, because $na_n = 0$ implies $a_n = 0$ for all $n \geq 1$. Hence:

Corollary 3.7. Every irreducible polynomial over a field of characteristic 0 is separable. Consequently every algebraic extension of a characteristic-0 field is separable.

In characteristic p , however,

$$\frac{d}{dx} x^p = px^{p-1} = 0.$$

More generally $f' = 0$ exactly when every exponent appearing in f is divisible by p , so $f(x) = g(x^p)$.

Worked example

Let $K = \mathbb{F}_p(t)$ and consider

$$f(x) = x^p - t.$$

It is irreducible over K (for example, t is not a p th power in K), but

$$f'(x) = px^{p-1} = 0.$$

If $\alpha^p = t$, then in a splitting field

$$x^p - t = x^p - \alpha^p = (x - \alpha)^p,$$

using the binomial theorem in characteristic p . Thus f has only one distinct root, of multiplicity p . This is the standard model of inseparability.

3.5 Counting embeddings

For a simple extension $K(\alpha)/K$, Proposition 3.2 says the number of K -embeddings into a field containing all conjugates equals the number of distinct roots of the minimal polynomial. Therefore it is at most $[K(\alpha) : K]$, with equality exactly when α is separable.

The same phenomenon persists for finite extensions.

Theorem 3.8 (Embedding count). *Let L/K be finite and let Ω be an algebraically closed field containing L . Then the number of K -embeddings $L \hookrightarrow \Omega$ is at most $[L : K]$. Equality holds if and only if L/K is separable.*

Proof idea and derivation. Choose a tower

$$K = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_r = L, \quad K_i = K_{i-1}(\alpha_i).$$

Suppose a K -embedding of K_{i-1} into Ω has already been chosen. To extend it across α_i , the image of α_i must be a root of the polynomial obtained by applying the existing embedding to the coefficients of $m_{\alpha_i, K_{i-1}}$. There are at most

$$[K_i : K_{i-1}]$$

such roots, and exactly that many if the relevant polynomial is separable. Multiplying the successive bounds yields

$$\# \text{Emb}_K(L, \Omega) \leq \prod_i [K_i : K_{i-1}] = [L : K].$$

For a finite separable extension every step can be taken separable and equality holds. Conversely, equality forces no loss of distinct roots at any step, hence separability. A fully formal proof is obtained by induction on the length of the tower using the extension-of-embeddings argument just described. $\square$

Common pitfall

“Algebraic” and “separable” answer different questions. Algebraic asks whether a finite polynomial relation exists. Separable asks whether that relation has the full number of distinct roots. In characteristic zero, separability is automatic, which is why it is easy to forget the distinction when first learning number theory.

3.6 Primitive elements

Theorem 3.9 (Primitive element theorem). *Every finite separable extension L/K is simple: there exists $\theta \in L$ such that*

$$L = K(\theta).$$

For the Chebotarev path, the theorem’s meaning matters more than memorizing every technical proof: finitely many separable generators can be encoded by one element. Here is the key two-generator argument when K is infinite.

Proof for two generators over an infinite base field. Let $L = K(\alpha, \beta)$ be finite separable. Work in an algebraic closure and list the distinct K -embeddings

$$\sigma_1, \dots, \sigma_n : L \hookrightarrow \Omega.$$

For $c \in K$, set $\theta_c = \alpha + c\beta$. If two embeddings agree on θ_c , then

$$\sigma_i(\alpha) + c\sigma_i(\beta) = \sigma_j(\alpha) + c\sigma_j(\beta).$$

If $\sigma_i(\beta) \neq \sigma_j(\beta)$, this determines at most one bad value

$$c = \frac{\sigma_j(\alpha) - \sigma_i(\alpha)}{\sigma_i(\beta) - \sigma_j(\beta)}.$$

If the denominator is zero, equality would force $\sigma_i(\alpha) = \sigma_j(\alpha)$, hence the embeddings agree on both generators and are equal. Therefore each pair of distinct embeddings excludes at most one c . There are only finitely many pairs, while K is infinite, so choose c avoiding all bad values. Then the restrictions of the n embeddings to $K(\theta_c)$ remain distinct.

By Theorem 3.8,

$$[K(\theta_c) : K] \geq n.$$

But separability gives $n = [L : K]$, while $K(\theta_c) \subseteq L$ gives

$$[K(\theta_c) : K] \leq [L : K].$$

Hence equality holds and $K(\theta_c) = L$. □

For finite base fields, the theorem follows by a different argument (and will be automatic from the cyclic structure of finite multiplicative groups in Chapter 8).

3.7 Definition recall drill

Without notes, reconstruct:

1. K -embedding and why it fixes K ;
2. separable polynomial, element, and extension;
3. repeated-root derivative criterion;
4. why characteristic zero implies separability;
5. why embeddings of $K(\alpha)$ correspond to roots of the minimal polynomial;
6. what the primitive element theorem is really saying.

3.8 Exercises

Exercise 3.1. List all $\mathbb{Q}$ -embeddings $\mathbb{Q}(\sqrt{2}) \hookrightarrow \mathbb{C}$ and prove there are exactly two.

Exercise 3.2. How many $\mathbb{Q}$ -embeddings $\mathbb{Q}(\sqrt[3]{2}) \hookrightarrow \mathbb{C}$ are there? Describe them explicitly. Why are these not automorphisms of $\mathbb{Q}(\sqrt[3]{2})$ except for one of them?

Exercise 3.3. Let K be a field. Determine when $x^n - 1$ is separable over K in terms of $\text{char } K$ and n .

Exercise 3.4. Let K have characteristic $p > 0$. Prove that $f'(x) = 0$ if and only if $f(x) = g(x^p)$ for some $g \in K[x]$.

Exercise 3.5. Let α be algebraic over K . Prove that every K -embedding $K(\alpha) \hookrightarrow \Omega$ sends α to a conjugate root, and conversely every conjugate root in Ω defines such an embedding.

Exercise 3.6. Give a complete proof that every irreducible polynomial over $\mathbb{Q}$ is separable.

Exercise 3.7. Show that $\mathbb{Q}(\sqrt{2}, \sqrt{3}) = \mathbb{Q}(\sqrt{2} + \sqrt{3})$ by explicitly recovering $\sqrt{2}$ and $\sqrt{3}$ from $\theta = \sqrt{2} + \sqrt{3}$.

Exercise 3.8. Let L/K be finite. Prove that $|\operatorname{Aut}_K(L)| \leq [L : K]$.

Exercise 3.9. Let $f, g \in K[x]$ be separable and coprime. Prove that fg is separable.

Exercise 3.10. Let $K = \mathbb{F}_p(t)$ and α satisfy $\alpha^p = t$. Prove that $K(\alpha)/K$ has degree p but only one K -embedding into an algebraic closure.

Chapter 4

Splitting fields and normality: containing all conjugates

4.1 Splitting fields

Definition 4.1 (Splitting field). Let $f \in K[x]$. A field extension L/K is a *splitting field of f over K* if

1. f factors into linear factors over L , and
2. L is generated over K by the roots of f .

Why this definition?

The first condition says “all roots are present.” The second prevents us from declaring an unnecessarily huge field to be the splitting field. Thus a splitting field is the smallest arena in which the polynomial can be seen completely.

Example 4.2. The splitting field of $x^2 - 2$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt{2})$. The splitting field of $x^3 - 2$ over $\mathbb{Q}$ is

$$\mathbb{Q}(\sqrt[3]{2}, \zeta_3), \quad \zeta_3 = e^{2\pi i/3},$$

because the three roots are $\sqrt[3]{2}$, $\zeta_3 \sqrt[3]{2}$, and $\zeta_3^2 \sqrt[3]{2}$.

Theorem 4.3 (Existence of splitting fields). *Every nonconstant polynomial $f \in K[x]$ has a splitting field over K .*

Proof. Induct on $\deg f$. If $\deg f = 1$, take K . Otherwise choose an irreducible factor g of f . Form

$$K_1 = K[x]/(g),$$

a field in which g , hence f , has at least one root α . In $K_1[x]$ write

$$f(x) = (x - \alpha)h(x)$$

with $\deg h < \deg f$. By induction h has a splitting field L over K_1 . Then f splits over L . Let E be the subfield of L generated over K by all roots of f . Then E is a splitting field by construction. $\square$

4.2 Uniqueness up to K -isomorphism

Splitting fields are not literally unique as subsets of some universal set, but they are unique in the only sense that matters algebraically: up to an isomorphism fixing K .

Lemma 4.4 (Extension lemma). *Let $\sigma : K \hookrightarrow K'$ be a field isomorphism, let $f \in K[x]$ be irreducible, and let $f^\sigma \in K'[x]$ be obtained by applying σ to coefficients. If α is a root of f and β a root of f^σ , then σ extends uniquely to an isomorphism*

$$K(\alpha) \xrightarrow{\sim} K'(\beta)$$

sending α to β .

Proof. Use quotient presentations

$$K(\alpha) \cong K[x]/(f), \quad K'(\beta) \cong K'[x]/(f^\sigma).$$

The coefficient map induced by σ sends (f) to (f^σ) , hence induces an isomorphism of quotients. The class of x maps to the class of x , i.e. α maps to β . $\square$

Theorem 4.5 (Uniqueness of splitting fields). *Any two splitting fields of $f \in K[x]$ are isomorphic by an isomorphism fixing K .*

Proof by extending root by root. Let L and L' be splitting fields. Start with the identity on K . Choose a root $\alpha \in L$ of an irreducible factor g of f . Since L' contains all roots of f , choose a root $\beta \in L'$ of g . Lemma 4.4 extends the identity to $K(\alpha) \cong K(\beta)$. Repeat: at each stage extend the current embedding by sending the next root in L to a compatible root in L' . Because both splitting fields are generated by finitely many roots, the process terminates with an embedding $L \hookrightarrow L'$. Symmetry and equal finite degrees (or repeating the construction in reverse) show it is onto. $\square$

4.3 Normal extensions

Definition 4.6 (Normal extension). An algebraic extension L/K is *normal* if every irreducible polynomial $f \in K[x]$ that has one root in L splits completely over L .

Why this definition?

Normality says: if the field contains one algebraic incarnation of a K -relation, it contains all its K -conjugates. So the field is closed under the symmetry that the base field cannot distinguish.

Example 4.7. $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ is normal because the conjugate of $\sqrt{2}$ is $-\sqrt{2}$, already present. But $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is not normal: it contains the real root of $x^3 - 2$ but not the nonreal roots $\zeta_3\sqrt[3]{2}$ and $\zeta_3^2\sqrt[3]{2}$.

Theorem 4.8 (Finite normal extensions are splitting fields). *For a finite extension L/K , the following are equivalent:*

- (i) L/K is normal;
- (ii) L is the splitting field over K of a finite family of polynomials in $K[x]$;
- (iii) every K -embedding $\sigma : L \hookrightarrow \bar{K}$ has image L .

Proof. (i) $\Rightarrow$ (ii): Since L/K is finite, choose generators $\alpha_1, \dots, \alpha_r$. Let m_i be the minimal polynomial of α_i over K . Normality implies each m_i splits in L . Hence the product $m_1 \cdots m_r$ splits in L , and its roots generate a field contained in L that contains all the α_i ; therefore it is L .

(ii) $\Rightarrow$ (iii): Suppose L is generated by all roots of polynomials $f_1, \dots, f_r \in K[x]$. A K -embedding σ sends every root of each f_i to another root of the same f_i . Thus σ permutes the generating set, so $\sigma(L) \subseteq L$. Since σ is injective and L/K is finite,

$$[\sigma(L) : K] = [L : K],$$

forcing $\sigma(L) = L$.

(iii) $\Rightarrow$ (i): Let $f \in K[x]$ be irreducible with a root $\alpha \in L$. Let β be any root of f in $\bar{K}$. Proposition 3.2 gives a K -embedding $K(\alpha) \hookrightarrow \bar{K}$ sending α to β . Theorem 3.3 extends it to a K -embedding $L \hookrightarrow \bar{K}$. By (iii), its image is L , so $\beta \in L$. Thus every conjugate root lies in L , and f splits. $\square$

Memory hook

Separable means “the conjugates are distinct.” Normal means “the conjugates are all inside.” Galois will mean both.

4.4 Normal closure

Definition 4.9 (Normal closure). For a finite extension E/K , a *normal closure* is a smallest normal extension L/K containing E .

If $E = K(\alpha_1, \dots, \alpha_r)$, take the splitting field of the product of the minimal polynomials of the α_i . This is a normal closure. In characteristic zero it is automatically separable and hence Galois.

Chebotarev connection

Chebotarev is stated for finite Galois extensions L/K . If your original arithmetic object naturally generates a nonnormal extension, you typically pass to its normal closure. The Galois group of that closure then acts on all conjugate roots at once.

4.5 Definition recall drill

State from memory:

1. splitting field and why the minimality clause matters;
2. normal extension and a nonnormal example;
3. the embedding characterization of normality;
4. normal closure.

Then explain in one sentence each: separability is about *multiplicity*; normality is about *location*.

4.6 Exercises

Exercise 4.1. Find a splitting field of $x^4 - 2$ over $\mathbb{Q}$ and compute its degree over $\mathbb{Q}$.

Exercise 4.2. Prove that every quadratic extension of a field of characteristic not 2 is normal and separable.

Exercise 4.3. Prove carefully that $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is not normal.

Exercise 4.4. Prove that the splitting field of $x^3 - 2$ over $\mathbb{Q}$ has degree 6.

Exercise 4.5. Let L_1/K and L_2/K be finite normal extensions inside a common algebraic closure. Prove that $L_1 \cap L_2$ is normal over K .

Exercise 4.6. Under the same hypotheses, prove that the compositum $L_1 L_2$ is normal over K .

Exercise 4.7. Let L/K be finite. Show directly that if every K -embedding $L \hookrightarrow \bar{K}$ maps L onto itself, then every conjugate over K of every $\alpha \in L$ belongs to L .

Exercise 4.8. Let $f \in K[x]$ be separable and let L be its splitting field. Prove that L/K is separable.

Exercise 4.9. Give an example in characteristic $p > 0$ of a finite normal extension that is not separable. Explain why it is normal despite inseparability.

Exercise 4.10. Determine the normal closure of $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ and explain why adjoining only one additional nonreal root automatically adjoins the third.

Chapter 5

Galois extensions and the fundamental correspondence

5.1 Automorphisms over the base field

Definition 5.1 (Galois group). For an extension L/K , define

$$\text{Gal}(L/K) = \text{Aut}_K(L),$$

the group of field automorphisms $\sigma : L \rightarrow L$ fixing every element of K .

Composition is the group law. The identity fixes everything; inverses exist because automorphisms are bijective.

Intuition

$\text{Gal}(L/K)$ measures the internal symmetries of L that are invisible from K . If K knows only an equation and not which conjugate root you chose, a Galois automorphism is allowed to exchange those indistinguishable choices while preserving all arithmetic relations.

Example 5.2. For $L = \mathbb{Q}(\sqrt{d})$ with d squarefree, every $\mathbb{Q}$ -automorphism is determined by the image of $\sqrt{d}$, which must be another root of $x^2 - d$. Hence

$$\text{Gal}(L/\mathbb{Q}) = \{1, \sigma\}, \quad \sigma(\sqrt{d}) = -\sqrt{d}.$$

Thus $\text{Gal}(L/\mathbb{Q}) \cong C_2$.

5.2 When does the group have the full degree?

By Theorem 3.8,

$$|\text{Gal}(L/K)| \leq [L : K].$$

Separability ensures there are $[L : K]$ embeddings into an algebraic closure. Normality ensures every such embedding lands back in L . Therefore:

Theorem 5.3 (Characterization of finite Galois extensions). *For a finite extension L/K , the following are equivalent:*

- (i) L/K is normal and separable;

- (ii) $|\text{Gal}(L/K)| = [L : K]$;
 (iii) K is exactly the fixed field of $\text{Gal}(L/K)$.

An extension satisfying these conditions is called Galois.

The implication (i) $\Rightarrow$ (ii) follows immediately from the preceding discussion. To establish the full theorem and the Galois correspondence rigorously, we need one structural result about finite groups of field automorphisms.

5.3 Fixed fields

Definition 5.4 (Fixed field). If G is a group of automorphisms of a field L , define

$$L^G = \{x \in L : \sigma(x) = x \text{ for every } \sigma \in G\}.$$

It is straightforward to check that L^G is a subfield: sums, products, negatives, and inverses of fixed elements remain fixed.

Why this definition?

A symmetry group hides whatever it moves and reveals whatever every symmetry leaves unchanged. The fixed field is the arithmetic information visible after we forget which Galois symmetry was applied.

5.4 Artin's independence lemma

We need to know that distinct field homomorphisms behave like linearly independent functions.

Lemma 5.5 (Artin independence of characters). Let $\sigma_1, \dots, \sigma_n : F \rightarrow L$ be distinct field homomorphisms. If

$$a_1\sigma_1(x) + \dots + a_n\sigma_n(x) = 0 \quad \text{for all } x \in F$$

with $a_i \in L$, then $a_1 = \dots = a_n = 0$.

Proof. Assume a nontrivial relation exists and choose one with the fewest nonzero coefficients. After renumbering, suppose $a_1 \neq 0$ and at least one other coefficient is nonzero. Because $\sigma_1 \neq \sigma_j$ for some j , choose $y \in F$ with $\sigma_1(y) \neq \sigma_j(y)$.

For arbitrary $x \in F$, the relation applied to xy gives

$$\sum_i a_i \sigma_i(x) \sigma_i(y) = 0.$$

Multiply the original relation at x by $\sigma_1(y)$ and subtract:

$$\sum_{i=2}^n a_i (\sigma_i(y) - \sigma_1(y)) \sigma_i(x) = 0.$$

The $i = 1$ coefficient has disappeared, while at least one remaining coefficient is nonzero. This is a shorter nontrivial relation, contradiction. $\square$

5.5 Artin's fixed-field theorem

Theorem 5.6 (Artin). *Let G be a finite group of automorphisms of a field L , and let $K = L^G$. Then*

$$[L : K] = |G|$$

and L/K is Galois with Galois group exactly G .

Proof. Write $G = \{\sigma_1, \dots, \sigma_n\}$, so $n = |G|$.

Step 1: prove $[L : K] \geq n$. If $[L : K] = m < n$, choose a K -basis $u_1, \dots, u_m$. Consider the $n \times m$ matrix

$$A = (\sigma_i(u_j)).$$

Since $m < n$, its rows are linearly dependent over L : there exist $a_i \in L$, not all zero, with

$$\sum_i a_i \sigma_i(u_j) = 0 \quad \text{for all } j.$$

Because every $x \in L$ is a K -linear combination of the u_j and every σ_i fixes K , the same relation holds for all $x \in L$, contradicting Artin independence. Hence $[L : K] \geq n$.

Step 2: prove $[L : K] \leq n$. Take any $\alpha \in L$. Form the orbit polynomial

$$P_\alpha(T) = \prod_{\beta \in G\alpha} (T - \beta),$$

where $G\alpha = \{\sigma(\alpha) : \sigma \in G\}$ uses distinct orbit elements. Every $\tau \in G$ permutes the factors, so it fixes every coefficient. Thus $P_\alpha \in K[T]$. Therefore α is algebraic over K of degree at most $|G\alpha| \leq n$.

To obtain a uniform bound on $[L : K]$, suppose for contradiction that there are $n + 1$ K -linearly independent elements $x_1, \dots, x_{n+1}$. The homogeneous system

$$\sum_{j=1}^{n+1} \sigma_i(x_j) c_j = 0 \quad (i = 1, \dots, n)$$

has a nonzero solution $(c_j) \in L^{n+1}$. Choose such a solution with the fewest nonzero c_j and normalize one nonzero coefficient to 1. Applying any $\tau \in G$ to the equations and permuting the rows shows $(\tau(c_j))$ is also a solution. Minimality and normalization force $\tau(c_j) = c_j$ for every j , hence $c_j \in K$. Then the equation corresponding to the identity automorphism gives

$$\sum_j x_j c_j = 0,$$

a nontrivial K -linear dependence, contradiction. Hence $[L : K] \leq n$.

Thus $[L : K] = n$. Since $G \subseteq \text{Gal}(L/K)$ and every finite extension has at most its degree many K -automorphisms,

$$|G| = n = [L : K] \geq |\text{Gal}(L/K)| \geq |G|,$$

so $G = \text{Gal}(L/K)$.

It remains to verify normality and separability without using the theorem we are in the process of proving. For any $\alpha \in L$, the orbit polynomial

$$P_\alpha(T) = \prod_{\beta \in G\alpha} (T - \beta) \in K[T]$$

has distinct roots, all lying in L , and satisfies $P_\alpha(\alpha) = 0$. The minimal polynomial $m_{\alpha,K}$ divides every polynomial in $K[T]$ that annihilates α , so $m_{\alpha,K} \mid P_\alpha$. Hence $m_{\alpha,K}$ also has distinct roots and all of them lie in L . Thus every element is separable over K , and every minimal polynomial of an element of L splits in L . Therefore L/K is separable and normal, hence Galois. $\square$

5.6 Completing the Galois characterization

If L/K is finite Galois and $G = \text{Gal}(L/K)$, then $K \subseteq L^G$. Artin's theorem gives

$$[L : L^G] = |G| = [L : K].$$

The tower law therefore forces $[L^G : K] = 1$, so $L^G = K$. Conversely, if $L^G = K$, Artin's theorem gives $[L : K] = |G|$, hence the full equivalence in Theorem 5.3.

5.7 The fundamental theorem of Galois theory

Theorem 5.7 (Fundamental theorem of Galois theory). *Let L/K be finite Galois with $G = \text{Gal}(L/K)$. There is an inclusion-reversing bijection*

$$\{\text{intermediate fields } K \subseteq E \subseteq L\} \longleftrightarrow \{\text{subgroups } H \leq G\}$$

given by

$$E \longmapsto \text{Gal}(L/E), \quad H \longmapsto L^H.$$

Moreover:

1. $[L : E] = |\text{Gal}(L/E)|$;
2. $[E : K] = [G : \text{Gal}(L/E)]$;
3. E/K is Galois if and only if $\text{Gal}(L/E) \trianglelefteq G$;
4. when E/K is Galois,

$$\text{Gal}(E/K) \cong G / \text{Gal}(L/E).$$

Proof. Let E be intermediate and put $H = \text{Gal}(L/E)$. Since L/K is normal and separable, L/E is also normal and separable: separability passes to larger base fields, and L remains a splitting field over E of the same family of polynomials. Hence L/E is Galois. By the fixed-field characterization, $L^H = E$.

Conversely, let $H \leq G$. Artin's theorem applied to H gives

$$[L : L^H] = |H|$$

and $\text{Gal}(L/L^H) = H$. Thus the two assignments are inverse.

The degree formulas follow from Artin's theorem and the tower law:

$$[E : K] = \frac{[L : K]}{[L : E]} = \frac{|G|}{|H|} = [G : H].$$

For normality, suppose E/K is Galois. If $\sigma \in G$ and $h \in H$, then for $x \in E$,

$$(\sigma h \sigma^{-1})(x) = \sigma(h(\sigma^{-1}(x))).$$

Because E/K is normal, every K -automorphism of L maps E to itself, so $\sigma^{-1}(x) \in E$; then h fixes it. Hence $\sigma h \sigma^{-1}$ fixes E , so it lies in H . Thus $H \trianglelefteq G$.

Conversely, if $H \trianglelefteq G$, then for $\sigma \in G$,

$$\sigma(L^H) = L^{\sigma H \sigma^{-1}} = L^H.$$

Now let $\tau : E = L^H \hookrightarrow \bar{K}$ be any K -embedding. By Theorem 3.3, τ extends to a K -embedding $\tilde{\tau} : L \hookrightarrow \bar{K}$. Because L/K is normal, $\tilde{\tau}(L) = L$, so $\tilde{\tau} \in G$. The displayed equality therefore gives

$$\tau(E) = \tilde{\tau}(E) = E.$$

Thus every K -embedding of E has image E , which gives normality by Theorem 4.8. Since E/K is separable as a subextension of the separable extension L/K , it is Galois.

Finally, restriction gives a homomorphism

$$\rho : G \rightarrow \text{Gal}(E/K), \quad \sigma \mapsto \sigma|_E.$$

Its kernel is exactly H . When E/K is Galois, every K -automorphism of E extends by Theorem 3.3 to a K -embedding of L into $\bar{K}$; normality of L/K makes the extension an element of G . Hence ρ is surjective. The first isomorphism theorem yields $G/H \cong \text{Gal}(E/K)$. $\square$

Intuition

The correspondence reverses inclusion because *more symmetries leave fewer elements fixed*. A large subgroup moves many elements, so its fixed field is small. A small subgroup imposes fewer invariance conditions, so its fixed field is large.

Memory hook

Fields go up when groups go down:

$$K \subseteq E \subseteq L \quad \Longleftrightarrow \quad G \supseteq H \supseteq \{1\}.$$

Normal subgroup $\leftrightarrow$ Galois intermediate extension.

5.8 Conjugation and conjugate subfields

For $H \leq G$ and $\sigma \in G$,

$$\sigma(L^H) = L^{\sigma H \sigma^{-1}}.$$

Thus conjugate subgroups correspond to conjugate intermediate fields. This is one of the first places where conjugacy is seen not as an arbitrary group-theoretic equivalence relation but as the natural ambiguity produced by changing coordinates inside a Galois extension.

Chebotarev connection

Later a prime $\mathfrak{p}$ of the base field will admit several primes $\mathfrak{P}$ above it. Choosing a different $\mathfrak{P}$ changes the associated Frobenius automorphism by conjugation. The group-theoretic mechanism is already visible here: changing a chosen object inside a transitive Galois orbit naturally conjugates its stabilizer or associated symmetry.

5.9 Definition recall drill

Reconstruct:

1. $\text{Gal}(L/K)$ and L^H ;
2. Galois $\Leftrightarrow$ finite, normal, separable;

3. why $|\text{Gal}(L/K)| \leq [L : K]$;
4. the two maps in the fundamental theorem;
5. why inclusion reverses;
6. normal subgroup $\leftrightarrow$ Galois intermediate field;
7. the quotient-group formula for $\text{Gal}(E/K)$.

5.10 Exercises

Exercise 5.1. Determine $\text{Gal}(\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q})$ explicitly and identify it up to isomorphism.

Exercise 5.2. List all intermediate fields of $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$ using the Galois correspondence.

Exercise 5.3. Compute the fixed field of complex conjugation in $\mathbb{C}$. Explain why Artin's fixed-field theorem actually implies $[\mathbb{C} : \mathbb{R}] = 2$ even if one has not first assumed that $\mathbb{C}/\mathbb{R}$ is finite, and then verify the degree directly from a basis.

Exercise 5.4. Let L/K be finite Galois with group G , and let $H \leq G$. Prove

$$[L^H : K] = [G : H].$$

Exercise 5.5. Use Galois theory to prove that every subgroup of index 2 is normal.

Exercise 5.6. In a finite Galois extension with group S_3 , determine which subgroup types correspond to intermediate extensions that are Galois over the base field.

Exercise 5.7. Let $L = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ and let σ fix $\sqrt{2}$ and send $\sqrt{3} \mapsto -\sqrt{3}$. Determine $L^{\langle \sigma \rangle}$ directly and verify the degree formula.

Exercise 5.8. Prove that $\text{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}))$ is trivial even though the extension has degree 3. Explain exactly which Galois condition fails.

Exercise 5.9. Suppose L_1/K and L_2/K are finite Galois inside a common algebraic closure. Prove that L_1L_2/K is Galois. Show that restriction gives an injective homomorphism

$$\text{Gal}(L_1L_2/K) \hookrightarrow \text{Gal}(L_1/K) \times \text{Gal}(L_2/K).$$

When is it an isomorphism?

Exercise 5.10. Let L/K be finite Galois and E intermediate. Prove directly that if $H = \text{Gal}(L/E)$ is normal in G , then every $\sigma \in G$ maps E to itself.

Chapter 6

Galois groups as permutation groups: concrete models

6.1 The group-action language you need

Definition 6.1 (Group action). An *action* of a group G on a set X is a rule $(g, x) \mapsto g \cdot x$ such that

$$1 \cdot x = x, \quad (gh) \cdot x = g \cdot (h \cdot x).$$

Equivalently, it is a group homomorphism

$$G \longrightarrow \text{Sym}(X),$$

where $\text{Sym}(X)$ is the group of bijections $X \rightarrow X$.

Why this definition?

An abstract group records multiplication but does not by itself say what the symmetries *do*. An action gives the group a stage on which to move objects. In Galois theory the stage will usually be a set of roots, primes, embeddings, or intermediate fields.

Definition 6.2 (Orbit, stabilizer, transitive, faithful). For $x \in X$,

$$Gx = \{g \cdot x : g \in G\}$$

is the *orbit* of x , and

$$G_x = \{g \in G : g \cdot x = x\}$$

is its *stabilizer*. The action is *transitive* if there is only one orbit. It is *faithful* if the only element acting trivially on every point is the identity.

Intuition

Orbit asks “where can symmetry move this object?” Stabilizer asks “which symmetries leave this object unchanged?” These are complementary: a large orbit means a small stabilizer.

Theorem 6.3 (Orbit-stabilizer). If G is finite, then

$$|Gx| = [G : G_x] = \frac{|G|}{|G_x|}.$$

Proof. Consider

$$\phi : G/G_x \longrightarrow Gx, \quad gG_x \longmapsto g \cdot x.$$

It is well defined: if $gG_x = hG_x$, then $h^{-1}g \in G_x$, so $(h^{-1}g) \cdot x = x$ and therefore $g \cdot x = h \cdot x$. It is surjective by definition of the orbit. If $g \cdot x = h \cdot x$, then $h^{-1}g \in G_x$, so $gG_x = hG_x$; hence it is injective. Thus the orbit is in bijection with the left cosets of the stabilizer. $\square$

Definition 6.4 (Conjugacy). $g, h \in G$ are *conjugate* if

$$h = xgx^{-1}$$

for some $x \in G$. The *conjugacy class* of g is

$$\text{Cl}(g) = \{xgx^{-1} : x \in G\}.$$

The *centralizer* of g is

$$C_G(g) = \{x \in G : xg = gx\}.$$

Conjugation is itself an action of G on G :

$$x \cdot g = xgx^{-1}.$$

Its orbit is the conjugacy class of g , while its stabilizer is $C_G(g)$. Orbit-stabilizer therefore gives

$$|\text{Cl}(g)| = [G : C_G(g)].$$

Memory hook

Conjugacy means “the same symmetry after changing coordinates inside the group.” This is why conjugacy, rather than equality, is the correct notion whenever an auxiliary choice changes the coordinate system.

6.2 Automorphisms permute roots

Let $f \in K[x]$ and suppose L/K is a splitting field of f . Every $\sigma \in \text{Gal}(L/K)$ sends a root α of f to another root because

$$f(\sigma(\alpha)) = \sigma(f(\alpha)) = 0.$$

Thus $G = \text{Gal}(L/K)$ acts on the root set X of f .

If f is separable, the action is faithful when L is generated by the roots: an automorphism fixing every root fixes the whole splitting field. Therefore

$$G \hookrightarrow S_X \cong S_n$$

when f has n roots.

Intuition

This is the origin of the classical phrase “Galois group of a polynomial.” The group is not invented by assigning arbitrary permutations to roots. It consists of exactly those root permutations compatible with every algebraic relation among the roots.

6.3 Irreducibility and transitivity

Proposition 6.5. *Let $f \in K[x]$ be irreducible and separable, and let L be its splitting field. Then $G = \text{Gal}(L/K)$ acts transitively on the roots of f .*

Proof. Let α and β be two roots. Since f is irreducible, it is the minimal polynomial of both. Proposition 3.2 gives a K -embedding

$$K(\alpha) \hookrightarrow L$$

sending α to β . Because L/K is normal, this embedding extends to a K -automorphism of L . Hence some $\sigma \in G$ sends α to β . $\square$

The converse also holds for a separable polynomial if we interpret transitivity on the full root set: if the Galois group acts transitively, all roots have the same minimal polynomial, so the polynomial is a power/scalar multiple of one irreducible factor; for a separable monic polynomial with all roots present once, this gives irreducibility.

6.4 Quadratic extensions

If $d \in K^\times$ is not a square and $\text{char } K \neq 2$, then

$$L = K(\sqrt{d})$$

is a degree-2 Galois extension. The nontrivial automorphism is

$$\sigma(a + b\sqrt{d}) = a - b\sqrt{d}.$$

The group is cyclic of order 2.

6.5 Biquadratic extensions

Let

$$L = \mathbb{Q}(\sqrt{2}, \sqrt{3}).$$

There are four automorphisms, obtained by independently choosing signs:

	$\sqrt{2}$	$\sqrt{3}$
1	$\sqrt{2}$	$\sqrt{3}$
σ	$-\sqrt{2}$	$\sqrt{3}$
τ	$\sqrt{2}$	$-\sqrt{3}$
$\sigma\tau$	$-\sqrt{2}$	$-\sqrt{3}$

Every nonidentity element has order 2, so

$$\text{Gal}(L/\mathbb{Q}) \cong C_2 \times C_2 =: V_4.$$

The three order-2 subgroups correspond to the three quadratic subfields

$$\mathbb{Q}(\sqrt{2}), \quad \mathbb{Q}(\sqrt{3}), \quad \mathbb{Q}(\sqrt{6}).$$

Worked example

Why is the third field $\mathbb{Q}(\sqrt{6})$? The subgroup $\langle \sigma\tau \rangle$ changes both signs, so it fixes the product $\sqrt{2}\sqrt{3} = \sqrt{6}$. Since its fixed field has degree 2 over $\mathbb{Q}$, containing $\mathbb{Q}(\sqrt{6})$ already forces equality.

6.6 The splitting field of $x^3 - 2$

Let

$$a = \sqrt[3]{2}, \quad \omega = e^{2\pi i/3},$$

so $\omega^2 + \omega + 1 = 0$. The splitting field is

$$L = \mathbb{Q}(a, \omega).$$

We already know $[\mathbb{Q}(a) : \mathbb{Q}] = 3$. Since $\mathbb{Q}(a) \subset \mathbb{R}$ but $\omega \notin \mathbb{R}$, we have $\omega \notin \mathbb{Q}(a)$, so

$$[L : \mathbb{Q}] = [L : \mathbb{Q}(a)][\mathbb{Q}(a) : \mathbb{Q}] = 2 \cdot 3 = 6.$$

The roots are

$$a, \quad \omega a, \quad \omega^2 a.$$

An automorphism is determined by the images of a and ω . We may define

$$r(a) = \omega a, \quad r(\omega) = \omega,$$

and

$$s(a) = a, \quad s(\omega) = \omega^2.$$

Then

$$r^3 = s^2 = 1,$$

and a direct calculation gives

$$srs = r^{-1}.$$

These are the standard relations for S_3 (equivalently the dihedral group of order 6), so

$$\text{Gal}(L/\mathbb{Q}) \cong S_3.$$

Worked example

As permutations of the three roots, r is the 3-cycle

$$(a \ \omega a \ \omega^2 a),$$

while complex conjugation s fixes a and swaps ωa with $\omega^2 a$. A 3-cycle and a transposition generate S_3 .

6.7 The Galois lattice for the S_3 example

The subgroup lattice of S_3 has:

- one subgroup A_3 of order 3, which is normal;
- three subgroups generated by transpositions, each of order 2 and not normal;
- the trivial subgroup and the whole group.

Correspondingly, $L/\mathbb{Q}$ has:

- one quadratic Galois subfield, fixed by A_3 ;
- three cubic non-Galois subfields, fixed by the order-2 subgroups.

The quadratic field is $\mathbb{Q}(\omega) = \mathbb{Q}(\sqrt{-3})$. One cubic field is $\mathbb{Q}(a)$; the others are its conjugates $\mathbb{Q}(\omega a)$ and $\mathbb{Q}(\omega^2 a)$ as embedded subfields of L .

6.8 Discriminants and parity of permutations

For a separable polynomial

$$f(x) = \prod_{i=1}^n (x - \alpha_i),$$

define the discriminant

$$\Delta(f) = \prod_{i < j} (\alpha_i - \alpha_j)^2.$$

Let

$$\delta = \prod_{i < j} (\alpha_i - \alpha_j),$$

so $\delta^2 = \Delta(f)$. If σ permutes the roots by $\pi \in S_n$, then the Vandermonde product changes by the sign of the permutation:

$$\sigma(\delta) = \text{sgn}(\pi)\delta.$$

Therefore:

Proposition 6.6. *Let $f \in K[x]$ be separable with splitting field L , and suppose $\text{char } K \neq 2$. Under the faithful permutation representation $G \leq S_n$,*

$$G \subseteq A_n \iff \Delta(f) \text{ is a square in } K.$$

Proof. If $G \subseteq A_n$, then every $\sigma \in G$ fixes δ , so $\delta \in L^G = K$. Hence $\Delta = \delta^2$ is a square in K .

Conversely, suppose $\Delta = d^2$ with $d \in K$. Then $\delta^2 = d^2$, so $(\delta - d)(\delta + d) = 0$ in the field L , giving $\delta = \pm d \in K$. Hence every $\sigma \in G$ fixes δ , so $\text{sgn}(\pi) = 1$ for every induced permutation. Thus $G \subseteq A_n$. $\square$

Chebotarev connection

In later arithmetic, factorization modulo a prime will reveal a cycle type in a Galois group. This chapter tells you what those cycles mean group-theoretically; Chapter 9 will show why finite-field Frobenius produces them.

6.9 Definition recall drill

Explain without notes:

1. why a Galois group acts faithfully on the roots of a polynomial whose splitting field is L ;
2. why irreducibility corresponds to transitivity;
3. the full V_4 example;
4. the full S_3 example for $x^3 - 2$;
5. why the discriminant detects whether the Galois group lies in A_n .

6.10 Exercises

Exercise 6.1. Verify explicitly that the automorphisms r, s above satisfy $srs = r^{-1}$.

Exercise 6.2. In the splitting field of $x^3 - 2$, identify the fixed field of A_3 and prove your answer using degrees.

Exercise 6.3. Identify the fixed field of the subgroup generated by complex conjugation. Why is it not Galois over $\mathbb{Q}$?

Exercise 6.4. Let $f \in K[x]$ be separable with splitting field L . Prove that if f is irreducible, then $\text{Gal}(L/K)$ acts transitively on its roots.

Exercise 6.5. Let $f \in \mathbb{Q}[x]$ be irreducible separable of degree 4. Explain why its Galois group, viewed in S_4 , must be a transitive subgroup. List at least three possible transitive subgroup isomorphism types.

Exercise 6.6. For $f(x) = x^3 + ax + b$, prove or verify from the root formula for the discriminant that

$$\Delta(f) = -4a^3 - 27b^2.$$

Then compute the discriminant of $x^3 - 2$ and use it to rule out A_3 as its Galois group.

Exercise 6.7. For each of the three nontrivial automorphisms of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, compute its fixed field directly.

Exercise 6.8. For the roots $\alpha_1, \dots, \alpha_n$ of a polynomial, explain why not every permutation of the roots need extend to a field automorphism. Give a concrete example where the Galois group is a proper subgroup of S_n .

Exercise 6.9. Draw the subgroup lattice of S_3 and, beside it, the reversed field lattice for the splitting field of $x^3 - 2$.

Exercise 6.10. Let $f \in \mathbb{Q}[x]$ be an irreducible separable cubic. Prove that its Galois group is A_3 if its discriminant is a square in $\mathbb{Q}$, and S_3 otherwise.

Chapter 7

Cyclotomic fields: arithmetic encoded as Galois symmetry

7.1 Roots of unity

An n th root of unity is a complex number ζ satisfying $\zeta^n = 1$. The roots are

$$1, e^{2\pi i/n}, e^{4\pi i/n}, \dots, e^{2\pi i(n-1)/n}.$$

A root is *primitive* if it has multiplicative order exactly n . Fix

$$\zeta_n = e^{2\pi i/n}.$$

Then the primitive roots are precisely

$$\zeta_n^a \quad (1 \leq a \leq n, \gcd(a, n) = 1).$$

There are $\varphi(n)$ of them.

7.2 Cyclotomic polynomials

Definition 7.1 (Cyclotomic polynomial). The n th cyclotomic polynomial is

$$\Phi_n(x) = \prod_{\substack{1 \leq a \leq n \\ \gcd(a, n) = 1}} (x - \zeta_n^a).$$

Every n th root of unity has a unique exact order d dividing n , so partitioning the roots by their exact multiplicative order gives

$$x^n - 1 = \prod_{d|n} \Phi_d(x).$$

Proposition 7.2. For every $n \geq 1$, $\Phi_n(x)$ is a monic polynomial in $\mathbb{Z}[x]$ of degree $\varphi(n)$.

Proof. The degree and monicity are immediate from the definition as the product over the $\varphi(n)$ primitive roots. We prove integrality by induction on n . For $n = 1$, $\Phi_1 = x - 1$. Assume $\Phi_d \in \mathbb{Z}[x]$ for every proper divisor d of n . Then

$$P(x) := \prod_{\substack{d|n \\ d < n}} \Phi_d(x)$$

is monic in $\mathbb{Z}[x]$, and the root-partition identity gives in $\mathbb{C}[x]$

$$x^n - 1 = P(x)\Phi_n(x).$$

Perform polynomial long division of $x^n - 1$ by the monic integer polynomial P . Because the divisor is monic, each division step subtracts an *integer* multiple of P , so both the quotient and remainder lie in $\mathbb{Z}[x]$. But the displayed equality over $\mathbb{C}[x]$ shows the remainder is actually zero and the quotient is Φ_n . Therefore $\Phi_n \in \mathbb{Z}[x]$. $\square$

Worked example

For $n = 6$,

$$x^6 - 1 = \Phi_1\Phi_2\Phi_3\Phi_6 = (x-1)(x+1)(x^2+x+1)\Phi_6,$$

so

$$\Phi_6(x) = x^2 - x + 1.$$

7.3 Irreducibility of cyclotomic polynomials

Theorem 7.3 (Cyclotomic irreducibility). $\Phi_n(x)$ is irreducible over $\mathbb{Q}$ for every $n \geq 1$.

Classical proof. Let $f \in \mathbb{Z}[x]$ be the monic irreducible factor of Φ_n having ζ_n as a root; Gauss's lemma allows us to work integrally. We show that for every prime $p \nmid n$, ζ_n^p is also a root of f .

Suppose not. Since ζ_n^p is primitive, it is a root of Φ_n , so it is a root of another factor $g \in \mathbb{Z}[x]$. Then $g(\zeta_n^p) = 0$, so ζ_n is a root of $g(x^p)$. Therefore $f(x)$ divides $g(x^p)$ in $\mathbb{Z}[x]$.

Reduce modulo p . In $\mathbb{F}_p[x]$,

$$\bar{g}(x^p) = \bar{g}(x)^p$$

by the freshman's dream. Hence $\bar{f}$ divides $\bar{g}^p$, so some irreducible factor of $\bar{f}$ also divides $\bar{g}$. Therefore $\bar{f}$ and $\bar{g}$ have a nonconstant common factor. Since $f\bar{g}$ divides Φ_n , it follows that the reduction of Φ_n has a repeated factor, hence a repeated root in an algebraic closure of $\mathbb{F}_p$.

But Φ_n divides $x^n - 1$, and because $p \nmid n$,

$$(x^n - 1)' = nx^{n-1}$$

is relatively prime to $x^n - 1$ modulo p . Thus $x^n - 1$, and hence Φ_n , is separable modulo p , contradiction.

So ζ_n^p is a root of f for every prime $p \nmid n$. By iteration, ζ_n^a is a root of f for every positive integer a coprime to n : factor a into primes, all of which are coprime to n . Hence f contains all primitive n th roots as roots. Therefore $f = \Phi_n$, proving irreducibility. $\square$

Remark 7.4. The reduction-mod- p step in this proof is an early hint of a recurring theme: arithmetic information about a characteristic-zero polynomial can be revealed by looking at its behavior modulo primes.

7.4 The Galois group of a cyclotomic field

Let

$$K_n = \mathbb{Q}(\zeta_n).$$

By irreducibility,

$$[K_n : \mathbb{Q}] = \varphi(n).$$

Since K_n contains every n th root of unity, it is the splitting field of $x^n - 1$, and characteristic zero gives separability. Thus $K_n/\mathbb{Q}$ is Galois.

Any automorphism is determined by ζ_n , and its image must be another primitive n th root. Conversely, every primitive root ζ_n^a is a root of the same minimal polynomial Φ_n , so Proposition 3.2 gives a $\mathbb{Q}$ -embedding

$$\sigma_a : \mathbb{Q}(\zeta_n) \hookrightarrow \mathbb{C}, \quad \sigma_a(\zeta_n) = \zeta_n^a, \quad a \in (\mathbb{Z}/n\mathbb{Z})^\times.$$

Its image is contained in $\mathbb{Q}(\zeta_n)$ because ζ_n^a lies there; equal finite degree then makes it an automorphism. Composition satisfies

$$\sigma_a \sigma_b(\zeta_n) = \zeta_n^{ab},$$

so

$$a \mapsto \sigma_a$$

is a group homomorphism

$$(\mathbb{Z}/n\mathbb{Z})^\times \longrightarrow \text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}).$$

It is injective, and both groups have $\varphi(n)$ elements. Hence:

Theorem 7.5.

$$\text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times.$$

Intuition

Cyclotomic Galois theory turns multiplication modulo n into automorphisms of a number field. The residue class a acts by raising a primitive root of unity to the a th power. This is the cleanest prototype for the Artin/Frobenius map that appears later.

7.5 Complex conjugation

Complex conjugation satisfies

$$\overline{\zeta_n} = \zeta_n^{-1}.$$

Under the isomorphism of Theorem 7.5, it corresponds to

$$-1 \in (\mathbb{Z}/n\mathbb{Z})^\times.$$

Its fixed field is the maximal real subfield

$$\mathbb{Q}(\zeta_n + \zeta_n^{-1}).$$

For $n > 2$ this subfield has degree $\varphi(n)/2$ over $\mathbb{Q}$.

7.6 Example: $n = 5$

Since

$$(\mathbb{Z}/5\mathbb{Z})^\times \cong C_4,$$

we get

$$\text{Gal}(\mathbb{Q}(\zeta_5)/\mathbb{Q}) \cong C_4.$$

There is a unique subgroup of order 2, so there is a unique quadratic intermediate field. The real subfield has degree 2, hence it is that field. In fact

$$\mathbb{Q}(\zeta_5 + \zeta_5^{-1}) = \mathbb{Q}(\sqrt{5}).$$

One can derive this from

$$\zeta_5 + \zeta_5^{-1} = 2 \cos(2\pi/5) = \frac{\sqrt{5} - 1}{2}.$$

7.7 Example: $n = 8$

$$(\mathbb{Z}/8\mathbb{Z})^\times = \{1, 3, 5, 7\} \cong C_2 \times C_2.$$

Thus $\mathbb{Q}(\zeta_8)/\mathbb{Q}$ is biquadratic. Since

$$\zeta_8 = \frac{1+i}{\sqrt{2}},$$

one finds

$$\mathbb{Q}(\zeta_8) = \mathbb{Q}(i, \sqrt{2}).$$

Its three quadratic subfields are

$$\mathbb{Q}(i), \quad \mathbb{Q}(\sqrt{2}), \quad \mathbb{Q}(\sqrt{-2}).$$

Chebotarev connection

For a rational prime $p \nmid n$, the later Frobenius automorphism in $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ will satisfy

$$\text{Frob}_p(\zeta_n) = \zeta_n^p.$$

Thus Frobenius will correspond exactly to the residue class $p \bmod n$. Chebotarev in this extension becomes the equidistribution of primes among invertible residue classes modulo n - the Galois-theoretic shadow of Dirichlet's theorem.

7.8 Definition recall drill

Reconstruct:

1. primitive n th root of unity;
2. $\Phi_n(x)$ and $x^n - 1 = \prod_{d|n} \Phi_d(x)$;
3. why $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n)$;
4. the map $(\mathbb{Z}/n\mathbb{Z})^\times \rightarrow \text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$;
5. why complex conjugation corresponds to -1 .

7.9 Exercises

Exercise 7.1. Compute $\Phi_n(x)$ for $n = 1, 2, 3, 4, 6, 8, 10, 12$.

Exercise 7.2. Compute $[\mathbb{Q}(\zeta_n) : \mathbb{Q}]$ for $n = 5, 7, 8, 9, 12, 15$.

Exercise 7.3. Describe $\text{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q})$ explicitly and determine its isomorphism type.

Exercise 7.4. Prove that the fixed field of complex conjugation in $\mathbb{Q}(\zeta_n)$ is $\mathbb{Q}(\zeta_n + \zeta_n^{-1})$ for $n > 2$.

Exercise 7.5. Derive algebraically, without quoting the cosine value, that $\mathbb{Q}(\zeta_5 + \zeta_5^{-1}) = \mathbb{Q}(\sqrt{5})$.

Exercise 7.6. Prove $\mathbb{Q}(\zeta_8) = \mathbb{Q}(i, \sqrt{2})$ and identify the automorphisms corresponding to $3, 5, 7 \in (\mathbb{Z}/8\mathbb{Z})^\times$ by their action on i and $\sqrt{2}$.

Exercise 7.7. Use the subgroup lattice of $(\mathbb{Z}/7\mathbb{Z})^\times$ to determine the number and degrees of proper nontrivial intermediate fields of $\mathbb{Q}(\zeta_7)/\mathbb{Q}$.

Exercise 7.8. Prove that every finite subgroup of the multiplicative group of a field is cyclic. Deduce that the group of n th roots of unity in a field is cyclic whenever it has n elements.

Exercise 7.9. Show that $\mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(\zeta_4) = \mathbb{Q}(i)$. Determine their intersection inside $\mathbb{C}$.

Exercise 7.10. Explain why every cyclotomic extension $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ is abelian. Why is this much stronger than merely being Galois?

Chapter 8

Finite fields: the canonical home of Frobenius

8.1 Characteristic and the prime field

If F is a field, the map

$$\mathbb{Z} \rightarrow F, \quad n \mapsto n \cdot 1_F$$

has kernel either (0) or (p) for a prime p . Thus the characteristic of a field is either 0 or prime. A finite field must have characteristic $p > 0$ and contains a copy of $\mathbb{F}_p$.

If $F/\mathbb{F}_p$ has finite dimension n , then as a vector space it has

$$|F| = p^n$$

elements. Hence every finite field has prime-power order.

8.2 The Frobenius map

Definition 8.1 (Frobenius endomorphism). For a field F of characteristic p , the map

$$\text{Fr}_p : F \rightarrow F, \quad x \mapsto x^p$$

is called the *Frobenius endomorphism*.

Why is it a field homomorphism? Multiplication is immediate:

$$(xy)^p = x^p y^p.$$

For addition, the binomial theorem gives

$$(x + y)^p = \sum_{k=0}^p \binom{p}{k} x^k y^{p-k}.$$

For $0 < k < p$, p divides $\binom{p}{k}$, so those coefficients vanish in characteristic p . Therefore

$$(x + y)^p = x^p + y^p.$$

It fixes $\mathbb{F}_p$ because Fermat's little theorem (or repeated addition) gives $a^p = a$ for $a \in \mathbb{F}_p$.

On any finite field, an injective self-map is surjective, so Frobenius is an automorphism.

Intuition

Frobenius is not an arbitrary automorphism chosen by hand. It is forced on every characteristic- p field by the arithmetic identity $(x + y)^p = x^p + y^p$. This canonicity is what makes it suitable for attaching a symmetry to a prime.

8.3 Roots of $x^q - x$

Let $q = p^n$. In any field with q elements, every nonzero element satisfies $x^{q-1} = 1$ because the multiplicative group has order $q - 1$; hence every element satisfies

$$x^q - x = 0.$$

The derivative is

$$qx^{q-1} - 1 = -1$$

in characteristic p , so the polynomial has no repeated roots. Thus its roots are exactly the q field elements.

Conversely, inside an algebraic closure $\overline{\mathbb{F}_p}$, define

$$F_q := \{x \in \overline{\mathbb{F}_p} : x^q = x\}.$$

The Frobenius identities show this set is closed under addition, multiplication, additive inverses, and inverses of nonzero elements. Since $x^q - x$ has exactly q distinct roots, F_q is a field with q elements.

This proves existence.

Theorem 8.2 (Existence and uniqueness of finite fields). *For every prime power $q = p^n$, there exists a field with q elements. Any two fields with q elements are isomorphic. Inside a fixed algebraic closure of $\mathbb{F}_p$, there is a unique subfield with q elements, namely the root set of $x^q - x$.*

Proof. Existence was constructed above. If F is any field with q elements, every element is a root of $x^q - x$, so an embedding of F into an algebraic closure identifies it with the full root set of that polynomial. Hence all such fields are isomorphic; inside a fixed algebraic closure, the root set is unique. $\square$

8.4 The Galois group of a finite field extension

Let $q = p^r$ and consider $\mathbb{F}_{q^n}/\mathbb{F}_q$. Define

$$\text{Fr}_q(x) = x^q.$$

This is the r th power of Fr_p , hence an $\mathbb{F}_q$ -automorphism. Iterating,

$$\text{Fr}_q^m(x) = x^{q^m}.$$

Every $x \in \mathbb{F}_{q^n}$ satisfies $x^{q^n} = x$, so

$$\text{Fr}_q^n = 1.$$

If $0 < m < n$ and $\text{Fr}_q^m = 1$ on all of $\mathbb{F}_{q^n}$, then all q^n elements would be roots of $x^{q^m} - x$, a polynomial of degree $q^m < q^n$, impossible. Thus the order of Frobenius is exactly n .

Since the extension degree is n , we have found n distinct automorphisms

$$1, \text{Fr}_q, \dots, \text{Fr}_q^{n-1}.$$

Therefore:

Theorem 8.3 (Finite-field Galois group).

$$\text{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q) = \langle \text{Fr}_q \rangle \cong C_n.$$

In particular every finite extension of finite fields is Galois and cyclic.

Memory hook

Finite fields are where Frobenius is simplest:

$$\text{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q) = \langle x \mapsto x^q \rangle.$$

Keep this formula available without thought. The number-field Frobenius in Phase 4 is designed so that its action on a residue field is exactly this automorphism.

8.5 Subfields of finite fields

By Galois correspondence, subfields of $\mathbb{F}_{q^n}$ containing $\mathbb{F}_q$ correspond to subgroups of the cyclic group C_n . There is exactly one subgroup of each order dividing n , so:

Theorem 8.4. *For each divisor $d \mid n$, $\mathbb{F}_{q^n}$ contains a unique subfield with q^d elements, namely $\mathbb{F}_{q^d}$. These are all the intermediate fields.*

Indeed the fixed field of Fr_q^d is the set satisfying $x^{q^d} = x$.

8.6 The multiplicative group is cyclic

Theorem 8.5. *The multiplicative group $\mathbb{F}_q^\times$ is cyclic of order $q - 1$.*

Proof. More generally, let G be a finite subgroup of the multiplicative group of a field. Let m be the exponent of G , the least common multiple of the orders of its elements. Every element of G is a root of $x^m - 1$, so $|G| \leq m$ because a degree- m polynomial has at most m roots. But by Lagrange, every element order divides $|G|$, so $m \mid |G|$ and hence $m \leq |G|$. Thus $m = |G|$.

It remains to see there is an element of order m . Write $m = \prod \ell_i^{a_i}$. For each prime power $\ell_i^{a_i}$, the definition of exponent guarantees some element has order divisible by $\ell_i^{a_i}$; taking a suitable power yields an element g_i of exact order $\ell_i^{a_i}$. Since the group is abelian and these orders are pairwise coprime, the product $\prod g_i$ has order $\prod \ell_i^{a_i} = m = |G|$. Hence G is cyclic. $\square$

8.7 Constructing finite fields as polynomial quotients

If $f \in \mathbb{F}_q[x]$ is irreducible of degree n , then

$$\mathbb{F}_q[x]/(f)$$

is a field with q^n elements, hence isomorphic to $\mathbb{F}_{q^n}$.

Worked example

Construct $\mathbb{F}_4$. Over $\mathbb{F}_2$, the polynomial

$$x^2 + x + 1$$

has no root, hence is irreducible. Let α be its residue class. Then

$$\mathbb{F}_4 = \mathbb{F}_2(\alpha) = \{0, 1, \alpha, \alpha + 1\},$$

with relation

$$\alpha^2 = \alpha + 1$$

because subtraction equals addition in characteristic 2. Frobenius is $x \mapsto x^2$:

$$\alpha \mapsto \alpha^2 = \alpha + 1, \quad \alpha + 1 \mapsto (\alpha + 1)^2 = \alpha^2 + 1 = \alpha.$$

Thus Frobenius swaps the two nontrivial conjugates and has order 2.

Worked example

Construct $\mathbb{F}_9$. The polynomial $x^2 + 1$ is irreducible over $\mathbb{F}_3$ because $-1 = 2$ is not a square modulo 3. Let $i^2 = -1$. Then

$$\mathbb{F}_9 = \mathbb{F}_3(i).$$

The $\mathbb{F}_3$ -Frobenius is

$$a + bi \mapsto a^3 + b^3 i^3 = a - bi,$$

so it plays exactly the role of complex conjugation in this quadratic finite-field extension.

8.8 Definition recall drill

State and derive:

1. why a finite field has order p^n ;
2. why $x \mapsto x^p$ is a homomorphism;
3. why the roots of $x^q - x$ form $\mathbb{F}_q$;
4. why $\text{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q)$ is cyclic of order n generated by $x \mapsto x^q$;
5. the classification of subfields;
6. why $\mathbb{F}_q^\times$ is cyclic.

8.9 Exercises

Exercise 8.1. Prove there is no field with exactly 6 elements.

Exercise 8.2. Find an irreducible cubic over $\mathbb{F}_2$ and use it to construct $\mathbb{F}_8$. Write down the Frobenius orbit of a root.

Exercise 8.3. List all subfields of $\mathbb{F}_{16}$ and justify that your list is complete.

Exercise 8.4. List all subfields of $\mathbb{F}_{64}$ containing $\mathbb{F}_2$. What are their degrees over $\mathbb{F}_2$?

Exercise 8.5. In $\mathbb{F}_{q^n}$, prove that the fixed field of Fr_q^d has $q^{\gcd(n,d)}$ elements.

Exercise 8.6. Prove directly that every $\mathbb{F}_q$ -automorphism of $\mathbb{F}_{q^n}$ is a power of Frobenius.

Exercise 8.7. Using $\mathbb{F}_9 = \mathbb{F}_3(i)$ with $i^2 = -1$, compute Fr_3 on all nine elements and verify its order is 2.

Exercise 8.8. Prove that $x^{q^n} - x$ is the product of all monic irreducible polynomials in $\mathbb{F}_q[x]$ whose degrees divide n , each appearing exactly once.

Exercise 8.9. Let $\alpha \in \overline{\mathbb{F}_q}$. Prove that the size of its Frobenius orbit

$$\alpha, \alpha^q, \alpha^{q^2}, \dots$$

is equal to the degree of its minimal polynomial over $\mathbb{F}_q$.

Exercise 8.10. Find a generator of $\mathbb{F}_7^\times$ and list its powers. Then find the order of every element of $\mathbb{F}_7^\times$.

Chapter 9

Frobenius, irreducible factors, and cycle type

This chapter is the conceptual payoff of Phase 2. It explains, entirely within finite-field Galois theory, why a polynomial's factorization pattern is encoded by a Frobenius permutation. In Phase 4 the same mechanism will be attached to prime ideals in number fields.

9.1 Frobenius acts on roots

Let $f \in \mathbb{F}_q[x]$ be separable, and let L be a splitting field. Because L is finite, $L = \mathbb{F}_{q^n}$ for some n , and

$$G = \text{Gal}(L/\mathbb{F}_q) = \langle \text{Fr}_q \rangle, \quad \text{Fr}_q(\alpha) = \alpha^q.$$

If α is a root of f , then

$$f(\alpha^q) = f(\alpha)^q = 0,$$

because the coefficients lie in $\mathbb{F}_q$ and are fixed by q th powers. Thus Frobenius permutes the roots.

9.2 Orbit length equals irreducible degree

Let $g \in \mathbb{F}_q[x]$ be monic irreducible of degree d and let α be one root. Then

$$\mathbb{F}_q(\alpha) \cong \mathbb{F}_{q^d}.$$

The Galois group is generated by Frobenius, and the conjugates of α are

$$\alpha, \alpha^q, \alpha^{q^2}, \dots, \alpha^{q^{d-1}}.$$

They are distinct, while $\alpha^{q^d} = \alpha$. Hence

$$g(x) = \prod_{j=0}^{d-1} (x - \alpha^{q^j}).$$

Theorem 9.1 (Factor degrees are Frobenius cycle lengths). *Let $f \in \mathbb{F}_q[x]$ be squarefree and factor into distinct monic irreducibles*

$$f = g_1 g_2 \cdots g_r, \quad \deg g_i = d_i.$$

On the roots of f in a splitting field, Fr_q has one cycle of length d_i for the roots of each g_i . Thus the cycle type of Frobenius is

$$(d_1)(d_2) \cdots (d_r).$$

Proof. Choose a root α_i of g_i . Because g_i is irreducible of degree d_i , the preceding orbit calculation shows that its roots are exactly

$$\alpha_i, \alpha_i^q, \dots, \alpha_i^{q^{d_i-1}},$$

and Frobenius cycles through them in order. Distinct irreducible factors have disjoint root sets because f is squarefree. Therefore the full permutation decomposes into exactly these cycles. $\square$

Intuition

An irreducible factor of degree d means: starting from one root, you need d Frobenius steps before returning. Factorization degree and orbit length are the same piece of information written in two languages.

9.3 Examples over prime fields

Worked example

Consider

$$f(x) = x^3 - 2 \in \mathbb{F}_5[x].$$

Test roots:

$$0^3 - 2 = 3, \quad 1^3 - 2 = 4, \quad 2^3 - 2 = 6 \equiv 1, \quad 3^3 - 2 = 25 \equiv 0.$$

So $x - 3$ is a factor. Polynomial division gives

$$x^3 - 2 = (x - 3)(x^2 + 3x + 4) \quad \text{in } \mathbb{F}_5[x].$$

The quadratic has discriminant

$$3^2 - 4 \cdot 4 = 9 - 16 = -7 \equiv 3 \pmod{5},$$

and 3 is not a square modulo 5, so the quadratic is irreducible. Thus the factorization type is $1 + 2$. Frobenius on the three roots therefore has cycle type $(1)(2)$, i.e. a transposition in S_3 .

Worked example

Modulo 7,

$$x^3 - 2$$

has no root: the cubes modulo 7 are 0, 1, 6. Hence the cubic is irreducible, so Frobenius acts as a 3-cycle on its roots.

Worked example

Modulo 31, $4^3 = 64 \equiv 2$, and because $31 \equiv 1 \pmod{3}$, the nonzero cube roots of unity lie in $\mathbb{F}_{31}$. Thus once one cube root of 2 exists, all three roots of $x^3 - 2$ lie in $\mathbb{F}_{31}$; the polynomial splits completely. Frobenius therefore acts trivially on its root set.

These three possibilities - identity, transposition, 3-cycle - are exactly the three conjugacy classes of S_3 .

9.4 Why cycle *type*, not a labeled permutation, is intrinsic

Suppose we label the roots $\alpha_1, \dots, \alpha_n$ and identify their permutation group with S_n . A different labeling is obtained by some $\tau \in S_n$. If Frobenius was represented by π in the first labeling, then in the new labeling it is represented by

$$\tau\pi\tau^{-1}.$$

Thus changing labels conjugates the permutation. The intrinsic data is therefore its conjugacy class, equivalently its cycle type.

Why this definition?

Conjugacy classes appear whenever an object is defined only after choosing coordinates or labels. Changing the choice conjugates the representing group element. A statement that is supposed to be choice-independent must therefore depend only on conjugacy-invariant information.

9.5 Factorization modulo primes as a preview of Chebotarev

Let $f \in \mathbb{Z}[x]$ be monic and separable over $\mathbb{Q}$, with splitting field $L/\mathbb{Q}$ and Galois group $G \leq S_n$. For a prime p at which reduction behaves well, reduce f modulo p . Its factor degrees define a cycle type. Phase 4 will show that, for unramified p , this is precisely the cycle type of the number-field Frobenius conjugacy class

$$\text{Frob}_p \subseteq G.$$

Chebotarev will then say that conjugacy classes appear with density proportional to their size.

For an S_3 cubic, this predicts:

conjugacy class	size	cycle type	factorization mod p
$\{1\}$	1	$(1)(1)(1)$	$1 + 1 + 1$
transpositions	3	$(1)(2)$	$1 + 2$
3-cycles	2	(3)	3

so the eventual densities should be

$$\frac{1}{6}, \quad \frac{1}{2}, \quad \frac{1}{3}.$$

At this stage this is only a preview; Chebotarev is what proves the global density statement.

9.6 Frobenius and cyclotomic fields: the residue-class prototype

Let $p \nmid n$. In a residue field containing an n th root of unity $\bar{\zeta}_n$, finite-field Frobenius acts by

$$\bar{\zeta}_n \mapsto \bar{\zeta}_n^p.$$

Compare this with the cyclotomic automorphism

$$\sigma_p(\zeta_n) = \zeta_n^p.$$

Thus the residue class $p \bmod n$ is already announcing which Galois automorphism Frobenius should be upstairs. This observation is one of the cleanest bridges from elementary congruences to algebraic number theory.

9.7 A reusable calculation protocol

When given a separable $f \in \mathbb{F}_q[x]$:

1. Factor f into irreducibles.
2. Record the degrees $d_1, \dots, d_r$.
3. Translate each degree d_i into one Frobenius cycle of length d_i .
4. If the polynomial came from a characteristic-zero polynomial with known Galois group, interpret that cycle type as a conjugacy-class constraint.

Conversely, if you know the Frobenius permutation, read off factor degrees from its cycles.

9.8 Definition recall drill

Without notes, explain:

1. why $\alpha \mapsto \alpha^q$ permutes roots of $f \in \mathbb{F}_q[x]$;
2. why a degree- d irreducible factor creates a d -cycle;
3. why relabeling roots conjugates the representing permutation;
4. how factorization modulo p foreshadows the number-field Frobenius class;
5. why $p \bmod n$ is related to Frobenius in a cyclotomic field.

9.9 Exercises

Exercise 9.1. Factor $x^3 - 2$ over $\mathbb{F}_5$, $\mathbb{F}_7$, and $\mathbb{F}_{31}$ and record the Frobenius cycle type in each case.

Exercise 9.2. Suppose a squarefree quartic over $\mathbb{F}_q$ factors as (a) four linears, (b) two irreducible quadratics, (c) linear times irreducible cubic, (d) irreducible quartic. Give the Frobenius cycle type in S_4 in each case.

Exercise 9.3. Let $\alpha \in \overline{\mathbb{F}_q}$ have minimal polynomial of degree d . Prove directly that

$$m_{\alpha, \mathbb{F}_q}(x) = \prod_{j=0}^{d-1} (x - \alpha^{q^j}).$$

Exercise 9.4. Let a permutation π describe a map on a root set under one labeling. Prove formally that after relabeling by τ , the new permutation is $\tau\pi\tau^{-1}$.

Exercise 9.5. Prove that two permutations in S_n are conjugate if and only if they have the same cycle type.

Exercise 9.6. Over $\mathbb{F}_3$, factor $x^4 - x$ and describe Frobenius on all its roots in a splitting field.

Exercise 9.7. Let $p \nmid n$. Show that the orbit length of a primitive n th root of unity under $x \mapsto x^p$ equals the multiplicative order of p in $(\mathbb{Z}/n\mathbb{Z})^\times$. Deduce that $\Phi_n(x)$ factors over $\mathbb{F}_p$ into irreducible factors all of degree $\text{ord}_n(p)$.

Exercise 9.8. Use the previous exercise to determine the degrees and number of irreducible factors of $\Phi_7(x)$ over $\mathbb{F}_2$ without explicitly factoring it.

Exercise 9.9. Suppose an irreducible cubic $f \in \mathbb{Q}[x]$ has nonsquare discriminant. If you find one good prime where f remains irreducible and another where it factors as linear times quadratic, what does this tell you about the observed cycle types? How does it agree with $\text{Gal}(f/\mathbb{Q}) \cong S_3$?

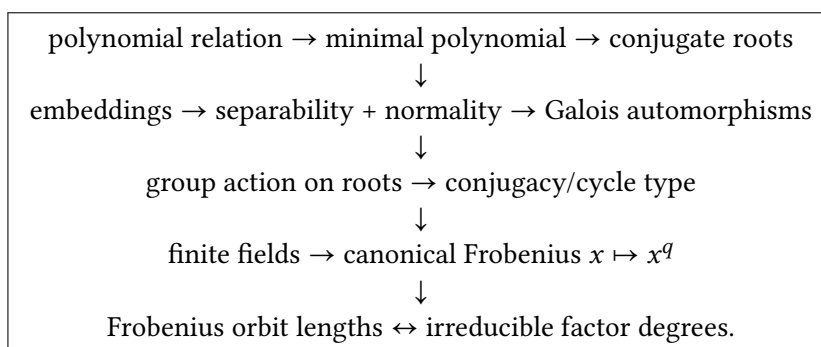
Exercise 9.10. For a squarefree polynomial over $\mathbb{F}_q$, prove that Frobenius fixes every root if and only if the polynomial splits completely over $\mathbb{F}_q$.

Chapter 10

Synthesis, study schedule, and readiness examination

10.1 The Phase 2 dependency map

At this point you should be able to reconstruct the following chain rather than merely read it:



The next phase changes the setting from arbitrary fields to number fields and their rings of integers. The crucial new issue will be that a rational or number-field prime can split into several prime ideals upstairs. But the residue fields at those primes are finite, so the Frobenius from this phase reappears exactly where we need it.

10.2 A four-week default schedule

Day	Work
1	Closed-book bridge diagnostic; Chapter 2 through the tower law.
2	Algebraic elements, minimal polynomials, simple extensions; Exercises 2.1–2.5.
3	Several generators and remaining Chapter 2 exercises; reconstruct Theorem 2.9.
4	Embeddings and conjugates; repeated-root criterion.
5	Separability and embedding count; exercises.
6	Primitive element theorem; reconstruct the proof idea without notes.
7	Weekly review: definitions from Chapters 2–3 from blank paper.
8	Splitting fields: existence and uniqueness.
9	Normality and equivalent criteria.
10	Galois groups, fixed fields, Artin independence.

11	Artin fixed-field theorem; do not rush this proof.
12	Fundamental theorem of Galois theory.
13	Galois-correspondence exercises.
14	Weekly review: draw the field/subgroup reversal from memory.
15	Permutation viewpoint and quadratic/biquadratic examples.
16	S_3 splitting field of $x^3 - 2$; draw the entire lattice.
17	Discriminant and parity; exercises.
18	Cyclotomic polynomials and roots of unity.
19	Cyclotomic irreducibility proof.
20	Cyclotomic Galois groups and examples $n = 5, 8$.
21	Weekly review: explain why $(\mathbb{Z}/n\mathbb{Z})^\times$ is a Galois group.
22	Finite-field existence/uniqueness and Frobenius.
23	$\text{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q)$ and subfields.
24	Explicit finite-field constructions; exercises.
25	Frobenius orbit/factorization theorem.
26	Factorization and conjugacy exercises.
27	Closed-book synthesis lecture; repair weak areas.
28	Final readiness exam, then mark using Appendix B.

10.3 How to study each theorem

For each theorem, make a four-line card:

1. **Statement:** what exactly is assumed and concluded?
2. **Why plausible:** what is the mental model?
3. **Proof skeleton:** what are the 2–5 decisive moves?
4. **Use later:** where does it enter Galois/Frobenius/Chebotarev?

A proof is not mastered when you can visually follow it. It is mastered when you can regenerate its skeleton and repair the technical details.

10.4 Final readiness examination

Take this closed-book. Allow 3–4 hours. Full solutions and marking guidance appear in Appendix B.

Exercise 10.1. Define: algebraic element, minimal polynomial, separable extension, normal extension, Galois extension, splitting field, fixed field, and Frobenius automorphism of $\mathbb{F}_{q^n}/\mathbb{F}_q$. For each definition add one sentence explaining its purpose.

Exercise 10.2. Prove the tower law and use it to show that if $[L : K] = 7$, there are no proper intermediate fields.

Exercise 10.3. Prove that if α is algebraic over K with minimal polynomial m of degree n , then

$$K(\alpha) \cong K[x]/(m)$$

and $1, \alpha, \dots, \alpha^{n-1}$ form a basis.

Exercise 10.4. Explain and prove the derivative criterion for separability. Why does it imply that every finite extension of $\mathbb{Q}$ is separable?

Exercise 10.5. State and prove the equivalence, for finite L/K , between normality and the property that every K -embedding $L \hookrightarrow \bar{K}$ has image L .

Exercise 10.6. State the fundamental theorem of Galois theory, including the normal-subgroup clause. Then apply it completely to $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$.

Exercise 10.7. Compute the splitting field and Galois group of $x^3 - 2$ over $\mathbb{Q}$. Describe the automorphisms as permutations of the roots and draw the subgroup/intermediate-field correspondence.

Exercise 10.8. Prove

$$\text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times,$$

assuming the irreducibility of Φ_n . Explain the role of complex conjugation under this isomorphism.

Exercise 10.9. Prove existence and uniqueness of the finite field $\mathbb{F}_q$ inside $\overline{\mathbb{F}_p}$ as the root set of $x^q - x$.

Exercise 10.10. Prove

$$\text{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q) = \langle x \mapsto x^q \rangle \cong C_n.$$

Classify the intermediate fields.

Exercise 10.11. Let $f \in \mathbb{F}_q[x]$ be squarefree with irreducible factor degrees 1, 2, 4. Describe Frobenius on its roots. Prove the general theorem relating factor degrees to Frobenius cycle lengths.

Exercise 10.12. Give a coherent ten-minute explanation of why *conjugacy classes* rather than individual permutations are the correct choice-independent objects when roots or primes above a base prime are not canonically labeled.

Exercise 10.13. For $p \nmid n$, explain why the action $\zeta_n \mapsto \zeta_n^p$ is the expected Frobenius action in the cyclotomic extension. Connect this to the residue class $p \bmod n$ and to the future statement of Chebotarev.

Exercise 10.14. Let $f \in \mathbb{Q}[x]$ be an irreducible separable cubic with nonsquare discriminant. Explain why its Galois group is S_3 . If a good reduction modulo p has factorization type $1 + 2$, identify the corresponding conjugacy class in S_3 .

Exercise 10.15. Write, from memory, the conceptual chain connecting the following words, with one or two sentences for each arrow:

minimal polynomial $\rightarrow$ conjugates $\rightarrow$ embeddings $\rightarrow$ Galois group
 $\rightarrow$ root action $\rightarrow$ Frobenius $\rightarrow$ factorization.

10.5 Readiness rubric

Score the final exam out of 100 using Appendix B. More importantly, classify each miss as *definition*, *proof architecture*, or *calculation*.

Mastery gate

Ready for Phase 3: score about 75% or higher, with no foundational definition error and with Problems 10.6, 10.7, 10.10, and 10.11 substantially correct.

Ready with one repair topic: score 65–75% and failures cluster in one area (for example cyclotomic details) while Galois correspondence and finite-field Frobenius are solid.

Do not move on yet: you still confuse separable/normal/Galois, cannot use the Galois correspondence in examples, or cannot explain why factor degrees are Frobenius cycle lengths. Those are load-bearing ideas for Phase 3 and Phase 4.

Appendix A

Complete solutions to all exercises

A.1 Solutions to the bridge diagnostic

Solution to Exercise 1.1

If f is irreducible, then every ideal of $F[x]$ containing (f) corresponds to a divisor of f . Since f is irreducible, the only such ideals are (f) and $(1) = F[x]$. Hence (f) is maximal, so $F[x]/(f)$ is a field.

Conversely, if $f = gh$ with g, h both nonconstant, then in the quotient the nonzero classes $\bar{g}$ and $\bar{h}$ satisfy $\bar{g}\bar{h} = \bar{f} = 0$. A field has no zero divisors, contradiction. Thus f must be irreducible.

Solution to Exercise 1.2

An ideal $\mathfrak{m}$ is maximal when there is no ideal strictly between $\mathfrak{m}$ and R . Ideals of $R/\mathfrak{m}$ correspond exactly to ideals of R containing $\mathfrak{m}$. Therefore $R/\mathfrak{m}$ has only the zero ideal and the whole ring exactly when $\mathfrak{m}$ is maximal. A commutative ring with 1 has only those two ideals exactly when every nonzero element is a unit, i.e. exactly when it is a field.

Intuitively, quotienting by a maximal ideal collapses as much of the ring as possible without collapsing 1 to 0; the result has no nontrivial ideal structure left, which is the ideal-theoretic signature of a field.

Solution to Exercise 1.3

Write $x = 2 + 5k$. Then $2 + 5k \equiv 3 \pmod{7}$, so $5k \equiv 1 \pmod{7}$. Since $5^{-1} \equiv 3 \pmod{7}$, $k \equiv 3 \pmod{7}$. Thus $x \equiv 2 + 15 = 17 \pmod{35}$.

The Chinese remainder theorem says that for coprime ideals (5) and (7) ,

$$\mathbb{Z}/(35) \xrightarrow{\sim} \mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/7\mathbb{Z}, \quad [x]_{35} \mapsto ([x]_5, [x]_7).$$

Hence each pair of residue classes has one and only one preimage modulo 35.

Solution to Exercise 1.4

$$R_{\mathfrak{p}} = S^{-1}R, \quad S = R \setminus \mathfrak{p}.$$

Its elements are fractions a/s with $a \in R$ and $s \notin \mathfrak{p}$. Every $s \notin \mathfrak{p}$ becomes a unit, with inverse $1/s$. More generally a/s is a unit precisely when $a \notin \mathfrak{p}$. Thus localization at $\mathfrak{p}$ makes everything away from $\mathfrak{p}$ invertible and leaves $\mathfrak{p}R_{\mathfrak{p}}$ as the unique maximal ideal.

Solution to Exercise 1.5

α is integral over A if it satisfies a monic equation

$$\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_0 = 0, \quad a_i \in A.$$

If K is a field and α is algebraic over K , it satisfies some nonzero polynomial $c_n x^n + \cdots + c_0$ with $c_n \neq 0$. Dividing by c_n makes the polynomial monic, so α is integral over K .

Solution to Exercise 1.6

For $v = \sum_i a_i v_i$, linearity forces

$$T(v) = \sum_i a_i T(v_i).$$

Thus the images of a basis determine T on every vector. A field automorphism of a finite extension L/K is in particular K -linear, so once its action on suitable generators/basis data is fixed, the entire map is forced. More sharply, an automorphism of $K(\alpha)$ is determined by the image of α , and that image must be a root of the minimal polynomial, yielding the degree bound on automorphisms.

A.2 Solutions to Chapter 2 exercises**Solution to Exercise 2.1**

For an intermediate field E , the tower law gives

$$p = [L : K] = [L : E][E : K].$$

Because p is prime and both factors are positive integers, one factor is 1. If $[E : K] = 1$, then $E = K$; if $[L : E] = 1$, then $E = L$. There is no proper intermediate field.

Solution to Exercise 2.2

Let $\alpha = \sqrt{2} + \sqrt{3}$. Then

$$\alpha^2 = 5 + 2\sqrt{6},$$

so

$$\alpha^2 - 5 = 2\sqrt{6}.$$

Squaring gives

$$(\alpha^2 - 5)^2 = 24,$$

hence

$$\alpha^4 - 10\alpha^2 + 1 = 0.$$

Thus α is a root of $f(x) = x^4 - 10x^2 + 1$.

To prove this is minimal, observe

$$\frac{1}{\alpha} = \sqrt{3} - \sqrt{2}$$

because $(\sqrt{3} + \sqrt{2})(\sqrt{3} - \sqrt{2}) = 1$. Hence

$$\sqrt{3} = \frac{1}{2}(\alpha + \alpha^{-1}), \quad \sqrt{2} = \frac{1}{2}(\alpha - \alpha^{-1}).$$

Therefore $\mathbb{Q}(\sqrt{2}, \sqrt{3}) \subseteq \mathbb{Q}(\alpha)$, while the reverse inclusion is obvious. Chapter 2 showed the biquadratic field has degree 4, so $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 4$. Hence the minimal polynomial has degree 4 and must be f .

Solution to Exercise 2.3

Let the monic minimal polynomial be

$$\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 = 0.$$

Because $\alpha \neq 0$ and the polynomial is irreducible, $a_0 \neq 0$ (otherwise x would divide it). Rearranging,

$$a_0 = -\alpha(\alpha^{n-1} + a_{n-1}\alpha^{n-2} + \cdots + a_1).$$

Thus

$$\alpha^{-1} = -a_0^{-1}(\alpha^{n-1} + a_{n-1}\alpha^{n-2} + \cdots + a_1) \in K[\alpha].$$

This is the concrete reason $K[\alpha]$ is already closed under inverses.

Solution to Exercise 2.4

$K(\alpha)/K$ is finite because α is algebraic. Since β is algebraic over K , the same polynomial shows it is algebraic over the larger field $K(\alpha)$, so $K(\alpha, \beta)/K(\alpha)$ is finite. The tower law gives finiteness over K .

Every element of a finite extension is algebraic by Proposition 2.14. Since $\alpha + \beta$, $\alpha\beta$, and α/β (when defined) all lie in $K(\alpha, \beta)$, they are algebraic over K .

Solution to Exercise 2.5

Because $K \subseteq K(\alpha) \subseteq L$, the tower law gives

$$[L : K] = [L : K(\alpha)][K(\alpha) : K].$$

Hence $[K(\alpha) : K]$ divides $[L : K]$.

Solution to Exercise 2.6

$x^3 - 2$ is irreducible over $\mathbb{Q}$ by the rational root test: a reducible cubic over a field has a linear factor, so it would have a rational root, and the only possible rational roots $\pm 1, \pm 2$ fail. Thus the minimal polynomial of $a = \sqrt[3]{2}$ has degree 3 and

$$[\mathbb{Q}(a) : \mathbb{Q}] = 3.$$

By the simple-extension theorem, $1, a, a^2$ form a basis. Since $a^2 = \sqrt[3]{4}$, each element has a unique expression

$$r + sa + ta^2, \quad r, s, t \in \mathbb{Q}.$$

Solution to Exercise 2.7

Evaluation $K[x] \rightarrow K[t]$, $f \mapsto f(t)$, is surjective by definition. Its kernel is zero because t is transcendental: no nonzero polynomial vanishes at t . Hence $K[x] \cong K[t]$.

But $K(t)$ is the fraction field of $K[t]$. For example $1/t \in K(t)$. If $1/t = f(t)$ for a polynomial f , then $tf(t) - 1 = 0$, giving a nonzero polynomial relation for t , impossible. Hence $K[t] \neq K(t)$.

Solution to Exercise 2.8

The tower law gives

$$4 = [L : E][E : K],$$

so $[E : K]$ can only be 1, 2, or 4. But divisibility is only a necessary condition. The tower law does not create an intermediate field. There are degree-4 extensions with no quadratic intermediate field; for example, in a Galois extension this can happen if the Galois group has no subgroup of the corresponding index (though every group of order 4 does, so one needs a nongalois degree-4 example). The logical point is that “degree divides” does not imply “subfield exists.”

Solution to Exercise 2.9

Suppose

$$a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6} = 0, \quad a, b, c, d \in \mathbb{Q}.$$

Rearrange:

$$a + b\sqrt{2} = -(c + d\sqrt{2})\sqrt{3}.$$

If $c + d\sqrt{2} \neq 0$, then

$$\sqrt{3} = -\frac{a + b\sqrt{2}}{c + d\sqrt{2}} \in \mathbb{Q}(\sqrt{2}),$$

contradicting the degree calculation that $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$. Hence $c + d\sqrt{2} = 0$, so $c = d = 0$ by independence of $1, \sqrt{2}$. Then $a + b\sqrt{2} = 0$, so $a = b = 0$.

Solution to Exercise 2.10

Divide f by the minimal polynomial m :

$$f = qm + r, \quad \deg r < \deg m.$$

Evaluate at α :

$$0 = f(\alpha) = q(\alpha)m(\alpha) + r(\alpha) = r(\alpha).$$

By minimality of m , no nonzero polynomial of lower degree can annihilate α . Therefore $r = 0$, so $m \mid f$.

A.3 Solutions to Chapter 3 exercises**Solution to Exercise 3.1**

A $\mathbb{Q}$ -embedding is determined by the image of $\sqrt{2}$, which must be a root of $x^2 - 2$. Thus the possibilities are

$$\sigma_+(a + b\sqrt{2}) = a + b\sqrt{2}, \quad \sigma_-(a + b\sqrt{2}) = a - b\sqrt{2}.$$

Both are field embeddings and there are no others.

Solution to Exercise 3.2

Let $a = \sqrt[3]{2}$. Its minimal polynomial $x^3 - 2$ has three distinct complex roots

$$a, \quad \omega a, \quad \omega^2 a.$$

Hence there are three $\mathbb{Q}$ -embeddings into $\mathbb{C}$, determined by sending a to one of these roots.

Only the identity has image inside the real field $\mathbb{Q}(a)$. The embeddings sending a to ωa or $\omega^2 a$ land in different conjugate subfields of $\mathbb{C}$, not in $\mathbb{Q}(a)$. Thus they are embeddings but not automorphisms of $\mathbb{Q}(a)$.

Solution to Exercise 3.3

For $f = x^n - 1$,

$$f' = nx^{n-1}.$$

If $\text{char } K$ does not divide n , then $n \neq 0$ in K , and any common root of f and f' would have to satisfy $x = 0$ from f' , but $f(0) = -1 \neq 0$. Hence f is separable.

If $\text{char } K = p$ divides n , write $n = pm$. Then

$$x^n - 1 = (x^m)^p - 1^p = (x^m - 1)^p,$$

so every root is repeated. Therefore $x^n - 1$ is separable exactly when $\text{char } K \nmid n$ (with characteristic 0 understood as never dividing n).

Solution to Exercise 3.4

Write

$$f(x) = \sum_{i=0}^N a_i x^i.$$

Then

$$f'(x) = \sum_{i=1}^N i a_i x^{i-1}.$$

In characteristic p , the coefficient $i a_i$ vanishes for nonzero a_i exactly when $p \mid i$. Thus $f' = 0$ exactly when every exponent with nonzero coefficient is a multiple of p . Then

$$f(x) = \sum_j a_{pj} x^{pj} = g(x^p), \quad g(y) = \sum_j a_{pj} y^j.$$

The converse is immediate by differentiating $g(x^p)$.

Solution to Exercise 3.5

If $\sigma : K(\alpha) \hookrightarrow \Omega$ fixes K , then

$$0 = \sigma(m(\alpha)) = m(\sigma(\alpha)),$$

so $\sigma(\alpha)$ is a root of m . Conversely, for any root β of m in Ω , the quotient construction

$$K(\alpha) \cong K[x]/(m) \rightarrow \Omega, \quad [x] \mapsto \beta$$

defines a K -embedding. Thus conjugate roots and embeddings are equivalent data for a simple algebraic extension.

Solution to Exercise 3.6

Let $f \in \mathbb{Q}[x]$ be irreducible. If f were inseparable, Proposition 3.6 would force $f' = 0$. But in characteristic zero, differentiating a nonconstant polynomial cannot give zero: if $f = \sum a_i x^i$ and $f' = 0$, then $i a_i = 0$ for each $i \geq 1$, so $a_i = 0$. Contradiction. Therefore every irreducible polynomial over $\mathbb{Q}$ is separable.

Solution to Exercise 3.7

Let $\theta = \sqrt{2} + \sqrt{3}$. As in Exercise 2.2,

$$\theta^{-1} = \sqrt{3} - \sqrt{2}.$$

Because $\theta \neq 0$, $\theta^{-1} \in \mathbb{Q}(\theta)$. Hence

$$\sqrt{3} = \frac{\theta + \theta^{-1}}{2}, \quad \sqrt{2} = \frac{\theta - \theta^{-1}}{2}.$$

Thus both generators of the biquadratic field lie in $\mathbb{Q}(\theta)$, so

$$\mathbb{Q}(\sqrt{2}, \sqrt{3}) = \mathbb{Q}(\theta).$$

Solution to Exercise 3.8

Every K -automorphism of L is a K -embedding of L into an algebraic closure. The embedding-count theorem gives at most $[L : K]$ such embeddings. Therefore

$$|\text{Aut}_K(L)| \leq [L : K].$$

Solution to Exercise 3.9

Because f and g are separable, neither has repeated roots. Because they are coprime, they have no common root in an algebraic closure. Therefore every root of fg occurs in exactly one of the two factors and with multiplicity one. Hence fg is separable. Equivalently,

$$\gcd(fg, (fg)') = \gcd(fg, f'g + fg') = 1.$$

Solution to Exercise 3.10

$f(x) = x^p - t$ is irreducible over $K = \mathbb{F}_p(t)$. One way to see this is the t -adic valuation: if $t = (a/b)^p$ in K , then the valuation of the right side would be divisible by p , whereas $v_t(t) = 1$; hence t is not a p th power, and $x^p - t$ has no smaller factor in this purely inseparable form. Thus $[K(\alpha) : K] = p$.

But

$$x^p - t = x^p - \alpha^p = (x - \alpha)^p$$

in an algebraic closure. The minimal polynomial has only one distinct root, so every K -embedding must send α to α . Hence there is exactly one embedding.

A.4 Solutions to Chapter 4 exercises**Solution to Exercise 4.1**

Let $a = \sqrt[4]{2}$. The roots are

$$a, -a, ia, -ia.$$

Thus the splitting field is

$$L = \mathbb{Q}(a, i).$$

$x^4 - 2$ is irreducible over $\mathbb{Q}$ by Eisenstein at 2, so $[\mathbb{Q}(a) : \mathbb{Q}] = 4$. Since $\mathbb{Q}(a) \subset \mathbb{R}$, it does not contain i , and i has degree 2 over $\mathbb{Q}(a)$ because $x^2 + 1$ has no real root. Therefore

$$[L : \mathbb{Q}] = 2 \cdot 4 = 8.$$

Solution to Exercise 4.2

Let $L = K(\alpha)$ with $[L : K] = 2$ and $\text{char } K \neq 2$. The minimal polynomial has degree 2, say

$$x^2 + bx + c.$$

If α is one root, the other root is $-b - \alpha$, which lies in L . Thus the minimal polynomial splits in L , and L is its splitting field, hence normal. Its derivative $2x + b$ cannot vanish at both roots; equivalently an irreducible quadratic in characteristic not 2 cannot have a repeated root. So the extension is separable, hence Galois.

Solution to Exercise 4.3

$a = \sqrt[3]{2}$ has irreducible minimal polynomial $x^3 - 2$. Its other roots are ωa and $\omega^2 a$, which are nonreal. But $\mathbb{Q}(a) \subset \mathbb{R}$, so they do not lie in $\mathbb{Q}(a)$. Therefore an irreducible polynomial over $\mathbb{Q}$ has one root in the field but does not split there. By definition the extension is not normal.

Solution to Exercise 4.4

The splitting field is $L = \mathbb{Q}(a, \omega)$. We have $[\mathbb{Q}(a) : \mathbb{Q}] = 3$. Since $\mathbb{Q}(a)$ is real and ω is not, $\omega \notin \mathbb{Q}(a)$. The polynomial $x^2 + x + 1$ shows $[\mathbb{Q}(a, \omega) : \mathbb{Q}(a)] \leq 2$, and nonmembership makes the degree exactly 2. Hence

$$[L : \mathbb{Q}] = 2 \cdot 3 = 6.$$

Solution to Exercise 4.5

Let $E = L_1 \cap L_2$. Take an irreducible $f \in K[x]$ with a root $\alpha \in E$. Then $\alpha \in L_1$ and $\alpha \in L_2$. Since both extensions are normal, all roots of f lie in each L_i , hence in their intersection E . Thus E/K is normal.

Solution to Exercise 4.6

Choose finite families of polynomials over K whose splitting fields are L_1 and L_2 . Then $L_1 L_2$ is generated by all roots of the union of these families, so it is the splitting field of their product. Hence $L_1 L_2 / K$ is normal.

Solution to Exercise 4.7

Let $\alpha \in L$ and let β be any K -conjugate of α , i.e. a root of its minimal polynomial. There is a K -embedding $K(\alpha) \hookrightarrow \bar{K}$ sending α to β . Extend it to a K -embedding $\sigma : L \hookrightarrow \bar{K}$. By hypothesis $\sigma(L) = L$, so $\beta = \sigma(\alpha) \in L$. Hence all conjugates of every element of L lie in L , which is the normality condition.

Solution to Exercise 4.8

Let L be generated by roots $\alpha_1, \dots, \alpha_r$ of a separable polynomial f . Each α_i is separable over K because its minimal polynomial divides f and therefore has distinct roots. A finite extension generated by finitely many separable elements is separable: build the tower and use the embedding-count characterization (the number of embeddings multiplies without loss at each separable step). Hence L/K is separable.

Solution to Exercise 4.9

Take $K = \mathbb{F}_p(t)$ and $L = K(\alpha)$ with $\alpha^p = t$. The minimal polynomial is $x^p - t = (x - \alpha)^p$ in an algebraic closure, so the extension is inseparable. It is nevertheless normal: the only distinct root of the minimal polynomial is α itself, so any irreducible polynomial over K having a root in L that arises from this purely inseparable extension has all its (single distinct) roots in L . Equivalently, every K -embedding $L \hookrightarrow \bar{K}$ has image L because there is only the identity embedding.

Solution to Exercise 4.10

The normal closure is the splitting field

$$L = \mathbb{Q}(a, \omega).$$

If we adjoin a and one nonreal root, say ωa , then

$$\omega = \frac{\omega a}{a}$$

also lies in the field. Then $\omega^2 a$ lies there as well. Thus one additional nonreal root supplies the primitive cube root of unity and hence the third root automatically.

A.5 Solutions to Chapter 5 exercises**Solution to Exercise 5.1**

Let $L = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Any $\mathbb{Q}$ -automorphism must send $\sqrt{2}$ to one of the roots $\pm\sqrt{2}$ of $x^2 - 2$, and independently send $\sqrt{3}$ to one of $\pm\sqrt{3}$. All four sign choices preserve the defining relations and extend to automorphisms:

$$\begin{aligned} 1, \quad \sigma : (\sqrt{2}, \sqrt{3}) &\mapsto (-\sqrt{2}, \sqrt{3}), \quad \tau : (\sqrt{2}, \sqrt{3}) \mapsto (\sqrt{2}, -\sqrt{3}), \\ \sigma\tau : (\sqrt{2}, \sqrt{3}) &\mapsto (-\sqrt{2}, -\sqrt{3}). \end{aligned}$$

Since $[L : \mathbb{Q}] = 4$, these are all automorphisms. Each nonidentity element has order 2, and the group is abelian, hence

$$\text{Gal}(L/\mathbb{Q}) \cong C_2 \times C_2 = V_4.$$

Solution to Exercise 5.2

The subgroups of V_4 are the whole group, the trivial subgroup, and the three order-2 subgroups. Reversing inclusion gives the fields

$$\mathbb{Q}, \quad L,$$

and three quadratic intermediate fields. Their identification is:

$$L^{\langle \sigma \rangle} = \mathbb{Q}(\sqrt{3}), \quad L^{\langle \tau \rangle} = \mathbb{Q}(\sqrt{2}), \quad L^{\langle \sigma\tau \rangle} = \mathbb{Q}(\sqrt{6}).$$

There are no others because the subgroup list is complete.

Solution to Exercise 5.3

Complex conjugation $c(z) = \bar{z}$ fixes exactly the real numbers, so $\mathbb{C}^{\{1, c\}} = \mathbb{R}$. Artin's fixed-field theorem actually applies to any field with a finite automorphism group; it then proves $[\mathbb{C} : \mathbb{R}] = 2$ directly for the finite group $\{1, c\}$. The point requiring care is that one must not assume in advance that $\mathbb{C}/\mathbb{R}$ is finite merely to use a theorem whose proof establishes finiteness. Directly, every complex number is uniquely $a + bi$ with $a, b \in \mathbb{R}$, so $\{1, i\}$ is an $\mathbb{R}$ -basis and $[\mathbb{C} : \mathbb{R}] = 2$.

Solution to Exercise 5.4

By Artin's theorem,

$$[L : L^H] = |H|.$$

Since L/K is Galois,

$$[L : K] = |G|.$$

The tower law gives

$$[L^H : K] = \frac{[L : K]}{[L : L^H]} = \frac{|G|}{|H|} = [G : H].$$

Solution to Exercise 5.5

Let $H \leq G$ have index 2. There are only two left cosets and two right cosets. Besides H itself, every element $g \notin H$ lies in the unique nontrivial left coset $gH = G \setminus H$ and also in the unique nontrivial right coset $Hg = G \setminus H$. Hence $gH = Hg$ for all g , so $H \trianglelefteq G$.

In Galois-theoretic language, an index-2 subgroup corresponds to a quadratic intermediate extension, and every separable quadratic extension is Galois; therefore the subgroup must be normal.

Solution to Exercise 5.6

The subgroups of S_3 are: S_3 , A_3 , three order-2 subgroups, and the identity. The normal subgroups are

$$\{1\}, \quad A_3, \quad S_3.$$

Thus the corresponding intermediate extensions Galois over the base are: the full field L/K , the quadratic field fixed by A_3 , and the base field itself. The cubic fields fixed by transposition subgroups are not Galois over K .

Solution to Exercise 5.7

Write an element as

$$x = a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6}.$$

Under σ fixing $\sqrt{2}$ and negating $\sqrt{3}$,

$$\sigma(x) = a + b\sqrt{2} - c\sqrt{3} - d\sqrt{6}.$$

Equality $\sigma(x) = x$ forces $c = d = 0$. Hence

$$L^{\langle \sigma \rangle} = \mathbb{Q}(\sqrt{2}).$$

The subgroup has order 2, so Artin's theorem gives $[L : \mathbb{Q}(\sqrt{2})] = 2$, consistent with $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$.

Solution to Exercise 5.8

Any $\mathbb{Q}$ -automorphism must send $a = \sqrt[3]{2}$ to another root of $x^3 - 2$ that lies in the same field $\mathbb{Q}(a)$. The other roots ωa and $\omega^2 a$ are nonreal, while $\mathbb{Q}(a) \subset \mathbb{R}$. Thus a must be fixed, so the automorphism is the identity. Therefore

$$|\text{Aut}_{\mathbb{Q}}(\mathbb{Q}(a))| = 1 < 3 = [\mathbb{Q}(a) : \mathbb{Q}].$$

The extension is separable (characteristic zero) but not normal.

Solution to Exercise 5.9

By Chapter 4, the compositum of finite normal extensions is normal. Separability is automatic if both are separable: L_1L_2 is generated by separable elements from L_1 and L_2 . Hence L_1L_2/K is Galois.

Restriction defines

$$\rho : \text{Gal}(L_1L_2/K) \rightarrow \text{Gal}(L_1/K) \times \text{Gal}(L_2/K), \quad \sigma \mapsto (\sigma|_{L_1}, \sigma|_{L_2}).$$

If both restrictions are identity, then σ fixes the field generated by L_1 and L_2 , namely L_1L_2 , so ρ is injective.

It is an isomorphism when $L_1 \cap L_2 = K$. Indeed then

$$[L_1L_2 : K] = [L_1 : K][L_2 : K],$$

so the injective map is between finite groups of equal order. More generally its image consists of pairs of automorphisms whose restrictions agree on $L_1 \cap L_2$.

Solution to Exercise 5.10

Let $E = L^H$ and suppose $H \trianglelefteq G$. For $x \in E$, $\sigma \in G$, and $h \in H$,

$$h(\sigma x) = \sigma(\sigma^{-1}h\sigma)(x).$$

Normality gives $\sigma^{-1}h\sigma \in H$, which fixes x . Thus

$$h(\sigma x) = \sigma x$$

for every $h \in H$, so $\sigma x \in L^H = E$. Therefore $\sigma(E) \subseteq E$, and applying the same argument to σ^{-1} gives equality.

A.6 Solutions to Chapter 6 exercises**Solution to Exercise 6.1**

Recall

$$r(a) = \omega a, \quad r(\omega) = \omega, \quad s(a) = a, \quad s(\omega) = \omega^2.$$

Check on the generators a, ω . First,

$$srs(a) = sr(a) = s(\omega a) = \omega^2 a = r^{-1}(a).$$

Also

$$srs(\omega) = sr(\omega^2) = s(\omega^2) = \omega = r^{-1}(\omega),$$

since r fixes ω . Thus $srs = r^{-1}$ on the generators, hence on all of L .

Solution to Exercise 6.2

$A_3 = \langle r \rangle$ has order 3, so its fixed field has degree

$$[S_3 : A_3] = 2$$

over $\mathbb{Q}$. Since $r(\omega) = \omega$, the field $\mathbb{Q}(\omega)$ is fixed by A_3 . It has degree 2 over $\mathbb{Q}$ because ω satisfies $x^2 + x + 1$. Hence

$$L^{A_3} = \mathbb{Q}(\omega) = \mathbb{Q}(\sqrt{-3}).$$

Solution to Exercise 6.3

Complex conjugation s fixes $a = \sqrt[3]{2}$, so $\mathbb{Q}(a) \subseteq L^{\langle s \rangle}$. The subgroup has order 2, hence its fixed field has degree $6/2 = 3$ over $\mathbb{Q}$. Since $\mathbb{Q}(a)$ already has degree 3, equality follows:

$$L^{\langle s \rangle} = \mathbb{Q}(a).$$

It is not Galois over $\mathbb{Q}$ because $\langle s \rangle$ is not normal in S_3 , or concretely because $x^3 - 2$ does not split in $\mathbb{Q}(a)$.

Solution to Exercise 6.4

Let α, β be roots of irreducible f . Since f is the minimal polynomial of α , Proposition 3.2 gives a K -embedding $K(\alpha) \rightarrow L$ sending α to β . Because L/K is a splitting field of a separable polynomial, it is Galois, and the embedding extends to a K -automorphism of L . Thus some group element carries α to β , proving transitivity.

Solution to Exercise 6.5

Irreducibility implies transitivity by Proposition 6.5. Transitive subgroups of S_4 include

$$C_4, \quad V_4, \quad D_4 \text{ (order 8)}, \quad A_4, \quad S_4.$$

For instance C_4 acts transitively via a 4-cycle; V_4 acts regularly on four points; and the larger groups are visibly transitive.

Solution to Exercise 6.6

Let the roots be r_1, r_2, r_3 . For the depressed cubic $x^3 + ax + b$, the elementary symmetric functions satisfy

$$r_1 + r_2 + r_3 = 0, \quad r_1 r_2 + r_1 r_3 + r_2 r_3 = a, \quad r_1 r_2 r_3 = -b.$$

Expanding

$$\prod_{i < j} (r_i - r_j)^2$$

and expressing the symmetric result in terms of the elementary symmetric functions gives

$$\Delta = -4a^3 - 27b^2.$$

This is a standard but mechanical symmetric-polynomial expansion; one efficient derivation also uses the resultant $\Delta = -\text{Res}(f, f')$ for a monic cubic.

For $x^3 - 2$, $a = 0$, $b = -2$, so

$$\Delta = -27 \cdot 4 = -108 = -3 \cdot 6^2,$$

which is not a square in $\mathbb{Q}$. Hence the Galois group is not contained in A_3 . Since the polynomial is irreducible, the group is transitive, so the only possibilities are A_3 or S_3 ; therefore it is S_3 .

Solution to Exercise 6.7

Using the notation from Chapter 6:

$$L^{\langle \sigma \rangle} = \mathbb{Q}(\sqrt{3}), \quad L^{\langle \tau \rangle} = \mathbb{Q}(\sqrt{2}), \quad L^{\langle \sigma\tau \rangle} = \mathbb{Q}(\sqrt{6}).$$

Directly, write $x = a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6}$ and equate x with its image under each sign change. The coefficients of the terms whose sign changes must vanish.

Solution to Exercise 6.8

A field automorphism must preserve *all* algebraic relations, not merely the equation $f(\alpha_i) = 0$. Thus a permutation of the roots extends only if it is compatible with sums, products, and every polynomial relation over K .

For example, the splitting field of $x^4 - 2$ has degree 8, so its Galois group has order 8, while S_4 has order 24. Hence most permutations of the four roots do not arise as automorphisms.

Solution to Exercise 6.9

The top subgroup S_3 corresponds to the bottom field $\mathbb{Q}$. The bottom subgroup $\{1\}$ corresponds to the top field L . The unique order-3 subgroup A_3 corresponds to the unique quadratic field $\mathbb{Q}(\omega)$. The three order-2 subgroups correspond to the three cubic conjugate subfields $\mathbb{Q}(a)$, $\mathbb{Q}(\omega a)$, and $\mathbb{Q}(\omega^2 a)$. Inclusion arrows reverse.

Solution to Exercise 6.10

For an irreducible cubic, transitivity forces the Galois group to be either A_3 or S_3 : these are the transitive subgroups of S_3 . Proposition 6.6 says the group lies in A_3 exactly when the discriminant is a square in $\mathbb{Q}$. Therefore square discriminant gives A_3 , nonsquare gives S_3 .

A.7 Solutions to Chapter 7 exercises**Solution to Exercise 7.1**

Use $x^n - 1 = \prod_{d|n} \Phi_d(x)$ recursively:

$$\begin{aligned}\Phi_1 &= x - 1, & \Phi_2 &= x + 1, & \Phi_3 &= x^2 + x + 1, \\ \Phi_4 &= x^2 + 1, & \Phi_6 &= x^2 - x + 1, \\ \Phi_8 &= x^4 + 1, & \Phi_{10} &= x^4 - x^3 + x^2 - x + 1, \\ \Phi_{12} &= x^4 - x^2 + 1.\end{aligned}$$

For example,

$$\Phi_8 = \frac{x^8 - 1}{\Phi_1 \Phi_2 \Phi_4} = \frac{x^8 - 1}{(x - 1)(x + 1)(x^2 + 1)} = x^4 + 1.$$

The others follow similarly.

Solution to Exercise 7.2

By cyclotomic irreducibility,

$$[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n).$$

Hence

$$\varphi(5) = 4, \quad \varphi(7) = 6, \quad \varphi(8) = 4, \quad \varphi(9) = 6, \quad \varphi(12) = 4, \quad \varphi(15) = 8.$$

Solution to Exercise 7.3

$$\text{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q}) \cong (\mathbb{Z}/7\mathbb{Z})^\times.$$

This group has order 6. It is cyclic: for example 3 has powers

$$3, 2, 6, 4, 5, 1 \pmod{7},$$

so it has order 6. Therefore the Galois group is C_6 , with automorphisms

$$\sigma_a(\zeta_7) = \zeta_7^a, \quad a = 1, 2, 3, 4, 5, 6.$$

Solution to Exercise 7.4

Let

$$t = \zeta_n + \zeta_n^{-1}.$$

Complex conjugation fixes t , so $\mathbb{Q}(t)$ lies in the fixed field. Conversely ζ_n satisfies

$$x^2 - tx + 1 = 0$$

over $\mathbb{Q}(t)$, hence

$$[\mathbb{Q}(\zeta_n) : \mathbb{Q}(t)] \leq 2.$$

For $n > 2$, ζ_n is nonreal, so it is not fixed by conjugation and therefore does not lie in the fixed field; the fixed field has index 2 by Galois correspondence. Since $\mathbb{Q}(t)$ is a fixed subfield with the correct degree, it equals the fixed field.

Solution to Exercise 7.5

Let

$$t = \zeta + \zeta^{-1}, \quad \zeta = \zeta_5.$$

From

$$1 + \zeta + \zeta^2 + \zeta^3 + \zeta^4 = 0,$$

divide by ζ^2 :

$$(\zeta^2 + \zeta^{-2}) + (\zeta + \zeta^{-1}) + 1 = 0.$$

But

$$\zeta^2 + \zeta^{-2} = t^2 - 2.$$

Thus

$$t^2 + t - 1 = 0.$$

Therefore $t = (-1 \pm \sqrt{5})/2$, so $\mathbb{Q}(t) = \mathbb{Q}(\sqrt{5})$. Since $t = 2 \cos(2\pi/5) > 0$, the positive value is $(\sqrt{5} - 1)/2$, but the field equality does not require choosing the sign.

Solution to Exercise 7.6

Since

$$\zeta_8 = e^{\pi i/4} = \frac{1+i}{\sqrt{2}},$$

we have $\zeta_8 \in \mathbb{Q}(i, \sqrt{2})$. Conversely,

$$\zeta_8^2 = i,$$

and

$$\zeta_8 + \zeta_8^{-1} = \sqrt{2},$$

so $i, \sqrt{2} \in \mathbb{Q}(\zeta_8)$. Hence the fields are equal.

For $a \in \{3, 5, 7\}$, $\sigma_a(\zeta_8) = \zeta_8^a$. Since $i = \zeta_8^2$,

$$\sigma_a(i) = \zeta_8^{2a} = i^a.$$

And since $\sqrt{2} = \zeta_8 + \zeta_8^{-1}$,

$$\sigma_a(\sqrt{2}) = \zeta_8^a + \zeta_8^{-a}.$$

This yields

a	i	$\sqrt{2}$
3	$-i$	$-\sqrt{2}$
5	i	$-\sqrt{2}$
7	$-i$	$\sqrt{2}$

(with $a = 1$ the identity).

Solution to Exercise 7.7

The group $(\mathbb{Z}/7\mathbb{Z})^\times \cong C_6$ has exactly one subgroup of each order dividing 6: orders 1, 2, 3, 6. Thus there are exactly two proper nontrivial intermediate fields. The order-3 subgroup has index 2, so its fixed field has degree 2 over $\mathbb{Q}$. The order-2 subgroup has index 3, so its fixed field has degree 3 over $\mathbb{Q}$.

Solution to Exercise 7.8

This is Theorem 8.5 in the more general setting of a finite subgroup G of a field's multiplicative group. Let m be the exponent of G . Every element is a root of $x^m - 1$, so $|G| \leq m$. Lagrange gives $m \mid |G|$, so $m = |G|$. Combining elements whose orders realize the prime-power factors of m produces an element of order m , hence a generator.

The set of n th roots of unity is a finite subgroup of the multiplicative group. If it has n elements, it is therefore cyclic of order n , generated by a primitive n th root.

Solution to Exercise 7.9

$\Phi_3 = x^2 + x + 1$, whose roots are

$$\frac{-1 \pm \sqrt{-3}}{2},$$

so $\mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$. Also $\zeta_4 = i$, so $\mathbb{Q}(\zeta_4) = \mathbb{Q}(i)$.

Their intersection is a subfield of each quadratic extension of $\mathbb{Q}$. A quadratic field has no proper intermediate field, so the intersection is either $\mathbb{Q}$ or the whole quadratic field. The two fields are unequal (for example one has discriminant -3 up to the field discriminant normalization, the other -4 ; more elementarily, equality would imply $\sqrt{3} \in \mathbb{Q}$ after comparing i and $i\sqrt{3}$). Hence the intersection is $\mathbb{Q}$.

Solution to Exercise 7.10

The isomorphism

$$\text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$$

identifies the Galois group with an abelian multiplicative group, so the extension is abelian. Being abelian is stronger than being Galois: Galois only requires the automorphism group to have full order

and the extension to be normal and separable; the group itself may be noncommutative, as in the S_3 splitting field of $x^3 - 2$.

A.8 Solutions to Chapter 8 exercises

Solution to Exercise 8.1

A finite field has order p^n for a prime p because it is a finite-dimensional vector space over its prime field $\mathbb{F}_p$. But 6 is not a prime power. Hence no field of order 6 exists.

Solution to Exercise 8.2

Take

$$f(x) = x^3 + x + 1 \in \mathbb{F}_2[x].$$

A cubic is reducible over a field if and only if it has a root. Testing 0 and 1 gives $f(0) = 1$ and $f(1) = 1$, so it is irreducible. Let α be its class in

$$\mathbb{F}_8 = \mathbb{F}_2[x]/(x^3 + x + 1).$$

Then $\alpha^3 = \alpha + 1$.

Frobenius is $x \mapsto x^2$. The orbit of α is

$$\alpha \mapsto \alpha^2 \mapsto \alpha^4 \mapsto \alpha^8 = \alpha.$$

The orbit has length 3, as expected from the minimal polynomial degree.

Solution to Exercise 8.3

$\mathbb{F}_{16} = \mathbb{F}_{2^4}$. Intermediate fields over $\mathbb{F}_2$ correspond to divisors of 4: 1, 2, 4. Thus the only subfields are

$$\mathbb{F}_2, \quad \mathbb{F}_4, \quad \mathbb{F}_{16}.$$

Uniqueness for each size follows from Theorem 8.4.

Solution to Exercise 8.4

$64 = 2^6$. The divisors of 6 are 1, 2, 3, 6. Hence the subfields containing $\mathbb{F}_2$ are

$$\mathbb{F}_2, \quad \mathbb{F}_4, \quad \mathbb{F}_8, \quad \mathbb{F}_{64},$$

with degrees 1, 2, 3, 6 over $\mathbb{F}_2$.

Solution to Exercise 8.5

Let $F = \mathbb{F}_{q^n}$ and $\sigma = \text{Fr}_q$. The subgroup generated by σ^d has order

$$\frac{n}{\gcd(n, d)}$$

in the cyclic group of order n . By Galois correspondence its fixed field has degree

$$\gcd(n, d)$$

over $\mathbb{F}_q$, hence contains

$$q^{\gcd(n, d)}$$

elements. Equivalently the fixed elements satisfy $x^{q^d} = x$, and their intersection with $\mathbb{F}_{q^n}$ is $\mathbb{F}_{q^{\gcd(n, d)}}$.

Solution to Exercise 8.6

We already have n distinct automorphisms

$$1, \text{Fr}_q, \dots, \text{Fr}_q^{n-1}$$

of the degree- n extension $\mathbb{F}_{q^n}/\mathbb{F}_q$. Any finite extension has at most its degree many base-field automorphisms. Therefore these n exhaust all automorphisms.

Solution to Exercise 8.7

Every element is $a + bi$ with $a, b \in \mathbb{F}_3$ and $i^2 = -1 = 2$. Frobenius gives

$$(a + bi)^3 = a^3 + b^3 i^3 = a + b(-i) = a - bi.$$

Thus it fixes the three elements of $\mathbb{F}_3$ and swaps each nonreal element with its conjugate:

$$i \leftrightarrow -i, \quad 1 + i \leftrightarrow 1 - i, \quad 2 + i \leftrightarrow 2 - i.$$

Applying twice gives the identity, so its order is 2.

Solution to Exercise 8.8

Let $h \in \mathbb{F}_q[x]$ be monic irreducible of degree d and let α be a root. Then $\mathbb{F}_q(\alpha) \cong \mathbb{F}_{q^d}$. The element α lies in $\mathbb{F}_{q^n}$ exactly when $\mathbb{F}_{q^d}$ is a subfield of $\mathbb{F}_{q^n}$, which by Theorem 8.4 occurs exactly when $d \mid n$. Thus h divides $x^{q^n} - x$ exactly when $d \mid n$.

The derivative of $x^{q^n} - x$ is -1 , so it is squarefree. Therefore each such monic irreducible factor appears exactly once. Since every root belongs to exactly one minimal polynomial, their product is the entire polynomial.

Solution to Exercise 8.9

Let $d = [\mathbb{F}_q(\alpha) : \mathbb{F}_q]$. Then $\mathbb{F}_q(\alpha) \cong \mathbb{F}_{q^d}$. Hence $\alpha^{q^d} = \alpha$, so the Frobenius orbit length divides d . If the orbit returned after $m < d$ steps, then $\alpha^{q^m} = \alpha$, so α would lie in the root field $\mathbb{F}_{q^m}$ and therefore have degree at most m , contradiction. Thus the orbit length is exactly d .

Solution to Exercise 8.10

Modulo 7, 3 is a generator:

$$3^1 = 3, \quad 3^2 = 2, \quad 3^3 = 6, \quad 3^4 = 4, \quad 3^5 = 5, \quad 3^6 = 1.$$

Thus $\mathbb{F}_7^\times \cong C_6$. The orders are:

$$\text{ord}(1) = 1, \quad \text{ord}(6) = 2,$$

$$\text{ord}(2) = \text{ord}(4) = 3,$$

$$\text{ord}(3) = \text{ord}(5) = 6.$$

A.9 Solutions to Chapter 9 exercises

Solution to Exercise 9.1

Over $\mathbb{F}_5$,

$$x^3 - 2 = (x - 3)(x^2 + 3x + 4),$$

with irreducible quadratic, so the cycle type is $1 + 2$, i.e. $(1)(2)$.

Over $\mathbb{F}_7$, there is no root because the cubes are $0, 1, 6$, so the cubic is irreducible and Frobenius is a 3-cycle.

Over $\mathbb{F}_{31}$, $4^3 \equiv 2$. Since $31 \equiv 1 \pmod{3}$, $\mathbb{F}_{31}$ contains the three cube roots of unity, so all three roots lie in $\mathbb{F}_{31}$ and the polynomial splits completely. Frobenius has cycle type $1 + 1 + 1$, i.e. the identity.

Solution to Exercise 9.2

By Theorem 9.1:

- (a) four linear factors $\leftrightarrow (1)(1)(1)(1)$;
- (b) two irreducible quadratics $\leftrightarrow (2)(2)$;
- (c) linear times irreducible cubic $\leftrightarrow (1)(3)$;
- (d) irreducible quartic $\leftrightarrow (4)$.

Solution to Exercise 9.3

Let d be the minimal-polynomial degree. The Frobenius orbit has exactly d distinct elements

$$\alpha, \alpha^q, \dots, \alpha^{q^{d-1}}.$$

Every one is a conjugate over $\mathbb{F}_q$ because Frobenius fixes the base field. Therefore the monic polynomial

$$P(x) = \prod_{j=0}^{d-1} (x - \alpha^{q^j})$$

has coefficients fixed by Frobenius: Frobenius cyclically permutes the factors. Hence the coefficients lie in the fixed field $\mathbb{F}_q$. P is monic of degree d and annihilates α , so by uniqueness of the monic minimal polynomial of degree d ,

$$P = m_{\alpha, \mathbb{F}_q}.$$

Solution to Exercise 9.4

Let the original labeling be a bijection $\ell : \{1, \dots, n\} \rightarrow X$. A permutation of the root set $F : X \rightarrow X$ is represented by

$$\pi = \ell^{-1} F \ell.$$

Relabel by $\ell' = \ell \tau^{-1}$ (so the new label i names the old point labeled $\tau^{-1}(i)$). Then the new representing permutation is

$$\pi' = (\ell')^{-1} F \ell' = \tau \ell^{-1} F \ell \tau^{-1} = \tau \pi \tau^{-1}.$$

Thus relabeling conjugates the permutation.

Solution to Exercise 9.5

If $\rho = \tau\pi\tau^{-1}$, then τ sends each cycle

$$(a_1 \ a_2 \ \dots \ a_r)$$

of π to the cycle

$$(\tau(a_1) \ \tau(a_2) \ \dots \ \tau(a_r))$$

of ρ . Hence conjugate permutations have the same cycle lengths.

Conversely, if π and ρ have the same cycle type, pair their cycles by length and choose a bijection τ sending the entries of each cycle of π in cyclic order to the entries of the paired cycle of ρ . Then $\tau\pi\tau^{-1} = \rho$.

Solution to Exercise 9.6

Over $\mathbb{F}_3$,

$$x^4 - x = x(x^3 - 1) = x(x - 1)(x^2 + x + 1).$$

But

$$x^2 + x + 1 = (x - 1)?$$

Check carefully: at $x = 1$, $1 + 1 + 1 = 3 = 0$, so

$$x^2 + x + 1 = (x - 1)(x + ?)$$

in $\mathbb{F}_3$. Dividing by $x - 1 = x + 2$ gives $x + 2$, so in fact

$$x^2 + x + 1 = (x - 1)(x + 2) = (x - 1)^2,$$

which reflects that $x^3 - 1 = (x - 1)^3$ in characteristic 3. Thus

$$x^4 - x = x(x - 1)^3$$

is not squarefree. The exercise intentionally exposes the squarefree hypothesis in Theorem 9.1. Its distinct roots are only 0 and 1, both in $\mathbb{F}_3$, and Frobenius fixes them. Multiplicity is not encoded by a permutation of distinct roots.

Solution to Exercise 9.7

Let ζ be a primitive n th root in an algebraic closure of $\mathbb{F}_p$. Frobenius iterates give

$$\zeta \mapsto \zeta^p \mapsto \zeta^{p^2} \mapsto \dots.$$

The first return after $d > 0$ occurs when

$$\zeta^{p^d} = \zeta,$$

equivalently

$$\zeta^{p^d - 1} = 1.$$

Since ζ has order n , this is equivalent to

$$p^d \equiv 1 \pmod{n}.$$

Thus $d = \text{ord}_n(p)$. Every primitive n th root has the same orbit length d , so the $\varphi(n)$ roots partition into $\varphi(n)/d$ Frobenius orbits. Each orbit is the root set of one irreducible factor, so Φ_n factors into $\varphi(n)/d$ irreducibles, each of degree d .

Solution to Exercise 9.8

Compute the order of 2 modulo 7:

$$2^1 = 2, \quad 2^2 = 4, \quad 2^3 = 8 \equiv 1 \pmod{7}.$$

Thus $\text{ord}_7(2) = 3$. Since $\varphi(7) = 6$, Φ_7 factors over $\mathbb{F}_2$ into

$$6/3 = 2$$

irreducible cubic factors.

Solution to Exercise 9.9

An irreducible reduction of a cubic gives factorization type 3, hence a Frobenius 3-cycle. A linear-times-quadratic reduction gives type 1 + 2, hence a transposition. These are exactly two of the three conjugacy-class types in S_3 . Together with the identity class (which corresponds to complete splitting), they fit the full cycle structure of S_3 . The observations do not by themselves prove a density theorem, but they are precisely the local signatures Chebotarev later distributes.

Solution to Exercise 9.10

If Frobenius fixes every root α , then $\alpha^q = \alpha$, so every root lies in the root set of $x^q - x$, namely $\mathbb{F}_q$. Hence the polynomial splits completely over $\mathbb{F}_q$. Conversely, if all roots lie in $\mathbb{F}_q$, then every root satisfies $\alpha^q = \alpha$, so Frobenius fixes each root.

Appendix B

Solutions and marking guide for the final readiness examination

Problem 10.1 - 10 points

Algebraic element: α is algebraic over K if some nonzero $f \in K[x]$ has $f(\alpha) = 0$. Purpose: it identifies elements governed by a finite polynomial relation over the base field.

Minimal polynomial: the unique monic irreducible polynomial $m_{\alpha,K} \in K[x]$ of least positive degree with $m(\alpha) = 0$. Purpose: it is the irreducible algebraic relation that contains all information about the simple extension $K(\alpha)/K$.

Separable extension: an algebraic extension in which every element has a minimal polynomial with distinct roots. Purpose: it ensures conjugates are genuinely distinct and that embedding counts reach their maximal value.

Normal extension: an algebraic extension L/K in which every irreducible $f \in K[x]$ having one root in L splits completely in L . Purpose: it ensures all K -conjugates of elements already present remain inside the field.

Galois extension: a finite extension that is normal and separable. Equivalently $|\text{Gal}(L/K)| = [L : K]$. Purpose: it is precisely the setting where the field is completely controlled by its full symmetry group.

Splitting field: the smallest extension generated by all roots of a polynomial (or family of polynomials). Purpose: it gives the minimal field in which the polynomial can be studied with all its conjugate roots present.

Fixed field: for $H \leq \text{Aut}(L)$,

$$L^H = \{x \in L : h(x) = x \ \forall h \in H\}.$$

Purpose: it recovers arithmetic information invariant under a group of symmetries.

Finite-field Frobenius: on $\mathbb{F}_{q^n}/\mathbb{F}_q$,

$$\text{Fr}_q(x) = x^q.$$

Purpose: it is the canonical generator of the finite-field Galois group and the model for Frobenius elements attached to primes later.

Award roughly one point per sound definition/purpose pair, with two extra points for precision across separable/normal/Galois.

Problem 10.2 - 6 points

Choose an E -basis $u_1, \dots, u_m$ of L and a K -basis $v_1, \dots, v_n$ of E . The products $u_i v_j$ span L over K by expanding coefficients first in the E -basis and then in the K -basis. They are independent because a K -linear relation can be regrouped as an E -linear relation among the u_i , then each coefficient is forced to vanish by independence of the v_j . Hence

$$[L : K] = mn = [L : E][E : K].$$

If $[L : K] = 7$, then

$$7 = [L : E][E : K].$$

Primality forces one factor to be 1, so $E = K$ or $E = L$.

Problem 10.3 - 7 points

Evaluation gives

$$\text{ev}_\alpha : K[x] \rightarrow K[\alpha], \quad f \mapsto f(\alpha).$$

Its kernel is (m) and its image is $K[\alpha]$. Therefore

$$K[x]/(m) \cong K[\alpha].$$

Since m is irreducible, (m) is maximal and the quotient is a field. Hence $K[\alpha] = K(\alpha)$. If $n = \deg m$, every class has a unique representative of degree $< n$, so

$$1, \alpha, \dots, \alpha^{n-1}$$

span. A linear dependence would be an annihilating polynomial of degree $< n$, contradicting minimality. Thus they form a basis.

Problem 10.4 - 6 points

A root α of f is repeated exactly when $(x - \alpha)^2 \mid f$. Writing $f = (x - \alpha)g$ gives

$$f' = g + (x - \alpha)g',$$

so $f'(\alpha) = g(\alpha)$. Thus α is repeated exactly when $f(\alpha) = f'(\alpha) = 0$. Hence f is separable iff $\gcd(f, f') = 1$.

If f is irreducible, inseparability forces $f \mid f'$, impossible by degree unless $f' = 0$. In characteristic zero a nonconstant polynomial never has zero derivative, so every irreducible polynomial over $\mathbb{Q}$ is separable. Therefore every finite extension of $\mathbb{Q}$ is separable.

Problem 10.5 - 8 points

Suppose L/K is normal. A K -embedding sends each element to a conjugate over K . For a finite set of generators, normality places all conjugates in L , so the image lies in L . Equal finite degree then makes the image all of L .

Conversely, suppose every K -embedding $L \hookrightarrow \bar{K}$ has image L . Let irreducible $f \in K[x]$ have a root $\alpha \in L$, and let β be any other root. There is a K -embedding $K(\alpha) \rightarrow \bar{K}$ taking α to β . Extend it to L . By hypothesis the image is L , hence $\beta \in L$. Therefore all roots lie in L and the extension is normal.

Full credit requires stating where extension of embeddings is used.

Problem 10.6 - 10 points

For finite Galois L/K with $G = \text{Gal}(L/K)$, intermediate fields E correspond inclusion-reversingly to subgroups H via

$$E \mapsto \text{Gal}(L/E), \quad H \mapsto L^H.$$

Moreover

$$[L : E] = |H|, \quad [E : K] = [G : H],$$

and E/K is Galois iff $H \trianglelefteq G$, in which case

$$\text{Gal}(E/K) \cong G/H.$$

For $L = \mathbb{Q}(\sqrt{2}, \sqrt{3})$,

$$G \cong V_4 = \{1, \sigma, \tau, \sigma\tau\}.$$

The three order-2 subgroups correspond to

$$\mathbb{Q}(\sqrt{3}), \quad \mathbb{Q}(\sqrt{2}), \quad \mathbb{Q}(\sqrt{6}),$$

respectively (depending on which sign change generates the subgroup). All subgroups are normal because V_4 is abelian, so all three quadratic intermediate extensions are Galois over $\mathbb{Q}$.

Problem 10.7 - 10 points

Let $a = \sqrt[3]{2}$ and $\omega = e^{2\pi i/3}$. The roots are

$$a, \omega a, \omega^2 a,$$

so the splitting field is $L = \mathbb{Q}(a, \omega)$. Since $x^3 - 2$ is irreducible, $[\mathbb{Q}(a) : \mathbb{Q}] = 3$. Since $\mathbb{Q}(a)$ is real and ω is not, adjoining ω contributes degree 2. Thus $[L : \mathbb{Q}] = 6$.

Define

$$r(a) = \omega a, \quad r(\omega) = \omega, \quad s(a) = a, \quad s(\omega) = \omega^2.$$

Then r is a 3-cycle on the roots, s a transposition, and they generate S_3 . Hence

$$\text{Gal}(L/\mathbb{Q}) \cong S_3.$$

The subgroup A_3 fixes $\mathbb{Q}(\omega)$; the three transposition subgroups fix the three cubic conjugate subfields. The trivial and full subgroups correspond to L and $\mathbb{Q}$.

Problem 10.8 - 6 points

Irreducibility gives

$$[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \deg \Phi_n = \varphi(n).$$

An automorphism must send ζ_n to another primitive n th root, so for $a \in (\mathbb{Z}/n\mathbb{Z})^\times$ define

$$\sigma_a(\zeta_n) = \zeta_n^a.$$

These are distinct automorphisms, and composition corresponds to multiplication modulo n . Therefore there is an injective homomorphism from $(\mathbb{Z}/n\mathbb{Z})^\times$ into the Galois group. Both groups have $\varphi(n)$ elements, so it is an isomorphism. Complex conjugation sends $\zeta_n \mapsto \zeta_n^{-1}$ and thus corresponds to $-1 \pmod n$.

Problem 10.9 - 6 points

Let $q = p^n$. In $\overline{\mathbb{F}_p}$ define

$$F = \{x : x^q = x\}.$$

The Frobenius identities imply closure under addition and multiplication:

$$(x + y)^q = x^q + y^q = x + y, \quad (xy)^q = x^q y^q = xy.$$

Negatives and nonzero inverses are also closed. The derivative of $x^q - x$ is -1 , so it has q distinct roots; hence $|F| = q$.

Any field with q elements has every nonzero element satisfying $x^{q-1} = 1$, so every element satisfies $x^q = x$. Thus after embedding in an algebraic closure it equals this root set, proving uniqueness up to isomorphism and uniqueness as a subfield of the fixed algebraic closure.

Problem 10.10 - 7 points

Frobenius

$$\sigma(x) = x^q$$

is an $\mathbb{F}_q$ -automorphism of $\mathbb{F}_{q^n}$. Its iterates are $x \mapsto x^{q^m}$. Since every element satisfies $x^{q^n} = x$, $\sigma^n = 1$. If $0 < m < n$ and $\sigma^m = 1$, then all q^n elements would be roots of $x^{q^m} - x$, a polynomial of smaller degree q^m , impossible. Thus σ has order n .

We have n distinct automorphisms of a degree- n extension, so they are all automorphisms:

$$\text{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q) = \langle \sigma \rangle \cong C_n.$$

Subgroups of C_n correspond to divisors $d \mid n$, yielding exactly the intermediate fields $\mathbb{F}_{q^d}$.

Problem 10.11 - 7 points

Degrees 1, 2, 4 give Frobenius cycle type

$$(1)(2)(4).$$

In general, for an irreducible factor g of degree d and a root α , the field $\mathbb{F}_q(\alpha)$ is $\mathbb{F}_{q^d}$. Its conjugates are

$$\alpha, \alpha^q, \dots, \alpha^{q^{d-1}},$$

with first return at step d . Thus Frobenius makes one d -cycle on the roots of g . Distinct squarefree factors have disjoint root sets, so the complete Frobenius permutation has one cycle whose length is the degree of each irreducible factor.

Problem 10.12 - 5 points

A satisfactory answer should articulate the coordinate-change principle. A root set has no canonical numbering. If an automorphism is represented by π under one labeling and the labels are changed by τ , its representation becomes

$$\tau \pi \tau^{-1}.$$

Thus the individual element of S_n depends on the labeling, while its conjugacy class does not. The same pattern will occur for primes $\mathfrak{P}$ above a base prime $\mathfrak{p}$: choosing a different prime above $\mathfrak{p}$ conjugates the associated Frobenius element. Therefore a base-prime invariant is naturally a conjugacy class.

Problem 10.13 - 5 points

In characteristic p , finite-field Frobenius raises elements to the p th power. Reducing an n th root of unity modulo a good prime therefore gives

$$\bar{\zeta}_n \mapsto \bar{\zeta}_n^p.$$

Upstairs in the cyclotomic Galois group, the automorphism

$$\sigma_p : \zeta_n \mapsto \zeta_n^p$$

corresponds under

$$\text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$$

to $p \bmod n$. Thus congruence classes of primes are Galois/Frobenius data. Chebotarev will say Frobenius classes are distributed according to group size; in this abelian cyclotomic case that becomes equidistribution among invertible residue classes, matching Dirichlet's theorem.

Problem 10.14 - 4 points

Irreducibility makes the Galois group a transitive subgroup of S_3 , so it is A_3 or S_3 . Nonsquare discriminant means the group is not contained in A_3 , hence it is S_3 . A factorization type $1 + 2$ modulo a good prime corresponds to a Frobenius permutation with one fixed point and one 2-cycle, i.e. the conjugacy class of transpositions.

Problem 10.15 - 3 points

One coherent chain is:

- A minimal polynomial is the irreducible relation defining an algebraic element over the base field.
- Its other roots are the conjugates: alternative algebraic realizations invisible to the base field.
- Base-field embeddings send an element precisely to such conjugates.
- When all conjugates are distinct and remain inside the extension (separable + normal), these embeddings become the full Galois group.
- The Galois group acts on root sets as a permutation group.
- Over finite fields there is a canonical Galois automorphism, Frobenius $x \mapsto x^q$.
- Frobenius orbits on roots have lengths equal to irreducible factor degrees, so its cycle type records factorization.

Interpreting the score

The numerical total is 100. Use it only together with the error types:

- **Definition errors** in separable/normal/Galois require repair before Phase 3.
- **Proof-architecture errors** in the tower law or simple extensions can often be repaired with two reconstruction sessions.
- **Computational errors** in a cyclotomic example are less serious if the general mechanism is sound.
- Failure on finite-field Frobenius or the factor-degree/cycle theorem is load-bearing and should be repaired before moving on.

Appendix C

Compact definition and theorem recall deck

Use this appendix for spaced recall. Cover the right-hand side and reconstruct it. Then add an example, a nonexample when applicable, and the reason the definition matters.

Prompt	Reconstruction target
Field extension L/K	K is a subfield of L ; L becomes a K -vector space.
Degree $[L : K]$	$\dim_K L$; finite degree measures the number of K -basis directions needed.
Algebraic element	Root of a nonzero polynomial in $K[x]$.
Minimal polynomial	Unique monic irreducible polynomial of least degree annihilating α .
Simple extension	$K(\alpha)$; if α is algebraic, $K(\alpha) \cong K[x]/(m_\alpha)$.
Tower law	$[L : K] = [L : E][E : K]$ for finite $K \subseteq E \subseteq L$.
K -embedding	Injective field homomorphism fixing K pointwise.
Conjugates of α	Roots of $m_{\alpha,K}$; possible images of α under K -embeddings.
Separable polynomial	Has no repeated roots in a splitting field; equivalent to $\gcd(f, f') = 1$.
Separable extension	Every element has separable minimal polynomial.
Splitting field	Smallest field generated by all roots of a polynomial/family.
Normal extension	Every irreducible polynomial over K with one root in L splits in L .
Galois extension	Finite, normal, and separable; equivalently automorphism count equals degree.
Galois group	$\text{Gal}(L/K) = \text{Aut}_K(L)$.
Fixed field L^H	Elements fixed by every $h \in H$.
Galois correspondence	$E \leftrightarrow \text{Gal}(L/E)$, $H \leftrightarrow L^H$, with inclusion reversed.
Normal subgroup clause	E/K Galois iff $\text{Gal}(L/E) \trianglelefteq \text{Gal}(L/K)$.
Cyclotomic polynomial	$\Phi_n = \prod_{(a,n)=1} (x - \zeta_n^a)$; irreducible over $\mathbb{Q}$.
Cyclotomic Galois group	$\text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$, $a : \zeta_n \mapsto \zeta_n^a$.
Finite field $\mathbb{F}_q$	Unique field of order $q = p^n$ up to isomorphism; roots of $x^q - x$ in $\overline{\mathbb{F}_p}$.
Frobenius	$\text{Fr}_q : x \mapsto x^q$ on extensions of $\mathbb{F}_q$.
Finite-field Galois group	$\text{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q) = \langle \text{Fr}_q \rangle \cong C_n$.
Finite-field subfields	$\mathbb{F}_{q^d} \subseteq \mathbb{F}_{q^n}$ exactly for $d \mid n$, uniquely.
Factor-cycle theorem	Irreducible factor degree d equals Frobenius orbit/cycle length d .

Why conjugacy?

Changing labels/choices conjugates a representing group element;
conjugacy class is choice-independent.

Appendix D

Proof reconstruction checklist

Before Phase 3, try to reconstruct these proofs from blank paper. A checkmark means you can recover the decisive idea and most details without looking.

1. [] Tower law.
2. [] $K(\alpha) \cong K[x]/(m_\alpha)$ and the power basis.
3. [] Minimal polynomials are irreducible.
4. [] Repeated-root/derivative criterion.
5. [] Embeddings of a simple extension correspond to conjugate roots.
6. [] Existence of splitting fields.
7. [] Normality via embeddings.
8. [] Artin independence lemma.
9. [] Fixed-field theorem $[L : L^G] = |G|$ for finite G .
10. [] Fundamental theorem of Galois theory, including the normal-subgroup clause.
11. [] $\text{Gal}(\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}) \cong V_4$ and its field lattice.
12. [] $\text{Gal}(\text{split}(x^3 - 2)/\mathbb{Q}) \cong S_3$ and its field lattice.
13. [] $\text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$ (assuming cyclotomic irreducibility).
14. [] Existence/uniqueness of $\mathbb{F}_q$ via $x^q - x$.
15. [] $\text{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q) = \langle x \mapsto x^q \rangle$.
16. [] Factor-degree $\leftrightarrow$ Frobenius-cycle theorem.

Chebotarev connection

If the last three proofs are fluent, you have reached the precise conceptual launching point for Phase 3 and Phase 4: prime ideals will produce finite residue fields; their finite-field Frobenius automorphisms will lift to Galois data; and factorization/splitting will be read from those automorphisms.

Closing perspective

The central achievement of Phase 2 is not memorizing the fundamental theorem of Galois theory. It is learning to see one phenomenon through several equivalent lenses:

$$\begin{array}{c} \text{algebraic relations} \longleftrightarrow \text{conjugate roots} \longleftrightarrow \text{field embeddings} \\ \longleftrightarrow \text{Galois symmetries} \longleftrightarrow \text{permutations.} \end{array}$$

Finite fields then add a canonical symmetry:

$$x \mapsto x^q.$$

Its orbits on roots are irreducible factors. This is the local algebraic mechanism Chebotarev eventually distributes across primes.

When that sentence feels like a compressed summary of mathematics you can reconstruct - rather than a slogan you recognize - Phase 2 has done its job.